

Revision of the Grade 8 course, and the entry diagnostic

Four things from last year decide this one: factorising, the discriminant, roots and fractions.

LESSON 1–2

2 × 40 MIN

ALGEBRA 9, POVTORENIE

IGX 1–6 REVIEW

LESSON CLOCK

8'

20'

20'

25'

7'

Why these four	8 min
Quadratic equations	20 min
Fractions and roots	20 min
The diagnostic	25 min
Targets	7 min

LEARNING OBJECTIVES

- Factorise a quadratic trinomial and solve a quadratic equation.
- Simplify an algebraic fraction and an expression with roots.
- Recognise which Grade 8 skill a Grade 9 question is really testing.
- Set a personal target from the diagnostic result.

TEXTBOOK

National	pp. 3–4
Cambridge	Core & Extended, pp. 1–130
Workbook	Diagnostic paper

01

Explanation

Why these four

Grade 9 opens with the quadratic **function**, and everything in the first chapter rests on four Grade 8 skills. If any one of them is shaky, the whole quarter is harder than it needs to be.

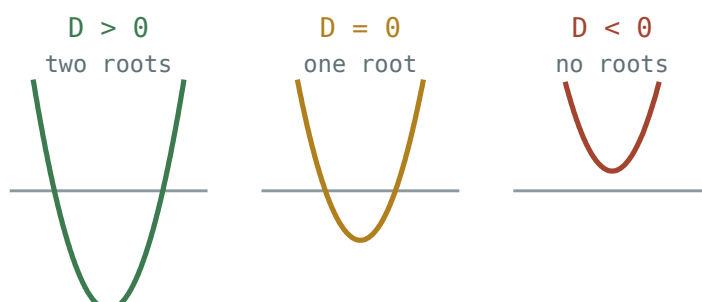
GRADE 8 SKILL	WHERE IT RETURNS
factorising $ax^2 + bx + c$	the roots, and where the parabola cuts Ox
the discriminant	how many times it cuts Ox at all
completing the square	the vertex, and the graph's position
algebraic fractions	the method of intervals, and the domain

ONE EQUATION, THREE READINGS

$x^2 - 5x + 6 = 0$ is a question about **numbers** ($x = 2$ or 3), about a **picture** (a parabola crossing Ox at 2 and 3) and about a **product** ($(x - 2)(x - 3) = 0$). Grade 9 uses all three at once, and the whole of this quarter is easier once you read them as the same thing.

Quadratic equations

$$ax^2 + bx + c = 0, D = b^2 - 4ac, x = \frac{-b \pm \sqrt{D}}{2a}$$



Three signs of D , three pictures.

D	ROOTS	THE PARABOLA
$D > 0$	two different	cuts Ox twice
$D = 0$	one repeated	touches Ox
$D < 0$	none real	misses Ox

VIETA FIRST, FORMULA SECOND

For $x^2 + px + q = 0$: $x_1 + x_2 = -p$ and $x_1 x_2 = q$. Two whole numbers whose sum is $-p$ and whose product is q are usually visible in a second — and when they are, the formula is a waste of a minute.

Fractions and roots

Cancel only factors. $\frac{x^2 - 9}{x + 3} = \frac{(x - 3)(x + 3)}{x + 3} = x - 3$, but only where $x \neq -3$ —

cancelling changes the domain, and Grade 9 will make that matter.

$$\sqrt{ab} = \sqrt{a} \cdot \sqrt{b} \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}} \sqrt{a^2} = |a|$$

$\sqrt{A^2}$ IS $|A|$, NOT A

$\sqrt{(-5)^2} = \sqrt{25} = 5$, not -5 . The arithmetic root is never negative, and this single fact is behind half the sign errors of the year.

The diagnostic

No	QUESTION	SKILL
1	Factorise $x^2 - 7x + 12$	factorising
2	Solve $2x^2 - 5x - 3 = 0$	the formula
3	Solve $x^2 + 4x + 7 = 0$	reading $D < 0$
4	Simplify $\frac{x^2 - 4}{x^2 + 4x + 4}$	fractions
5	Simplify $\sqrt{50} - \sqrt{18}$	roots
6	Write $x^2 - 6x + 5$ as $(x - m)^2 + n$	completing the square
7	Find $x_1 + x_2$ and x_1x_2 for $x^2 - 8x + 15 = 0$	Vieta
8	Solve $\frac{1}{x} + \frac{1}{x+1} = \frac{3}{2}$	fractional equations

WHAT THE SCORE MEANS

Do not count the total. Count how many of the four skills you lost a mark on. One weak skill is about twenty minutes of practice; three weak skills is why Chapter I will feel impossible, and is worth fixing this week rather than in December.

02 Worked examples

EXAMPLE 1 Solve $2x^2 - 5x - 3 = 0$.

$$① \quad D = 25 + 24 = 49$$

$$② \quad \sqrt{D} = 7$$

$$③ \quad x = \frac{5 \pm 7}{4}$$

$$④ \quad x = 3 \text{ or } x = -0.5$$

ANSWER

$$x = 3 \text{ and } x = -0.5$$

EXAMPLE 2 Simplify $\frac{x^2 - 4}{x^2 + 4x + 4}$.

$$① \quad \frac{(x-2)(x+2)}{(x+2)^2}$$

Factorise both.

$$② \quad = \frac{x-2}{x+2}$$

$$③ \quad x \neq -2$$

Say so — the domain did not change.

ANSWER

$$\frac{x-2}{x+2}, x \neq -2$$

EXAMPLE 3 Write $x^2 - 6x + 5$ in the form $(x - m)^2 + n$.

① $x^2 - 6x = (x - 3)^2 - 9$

Half of 6, squared.

② $(x - 3)^2 - 9 + 5$

③ $= (x - 3)^2 - 4$

The vertex will be $(3, -4)$.

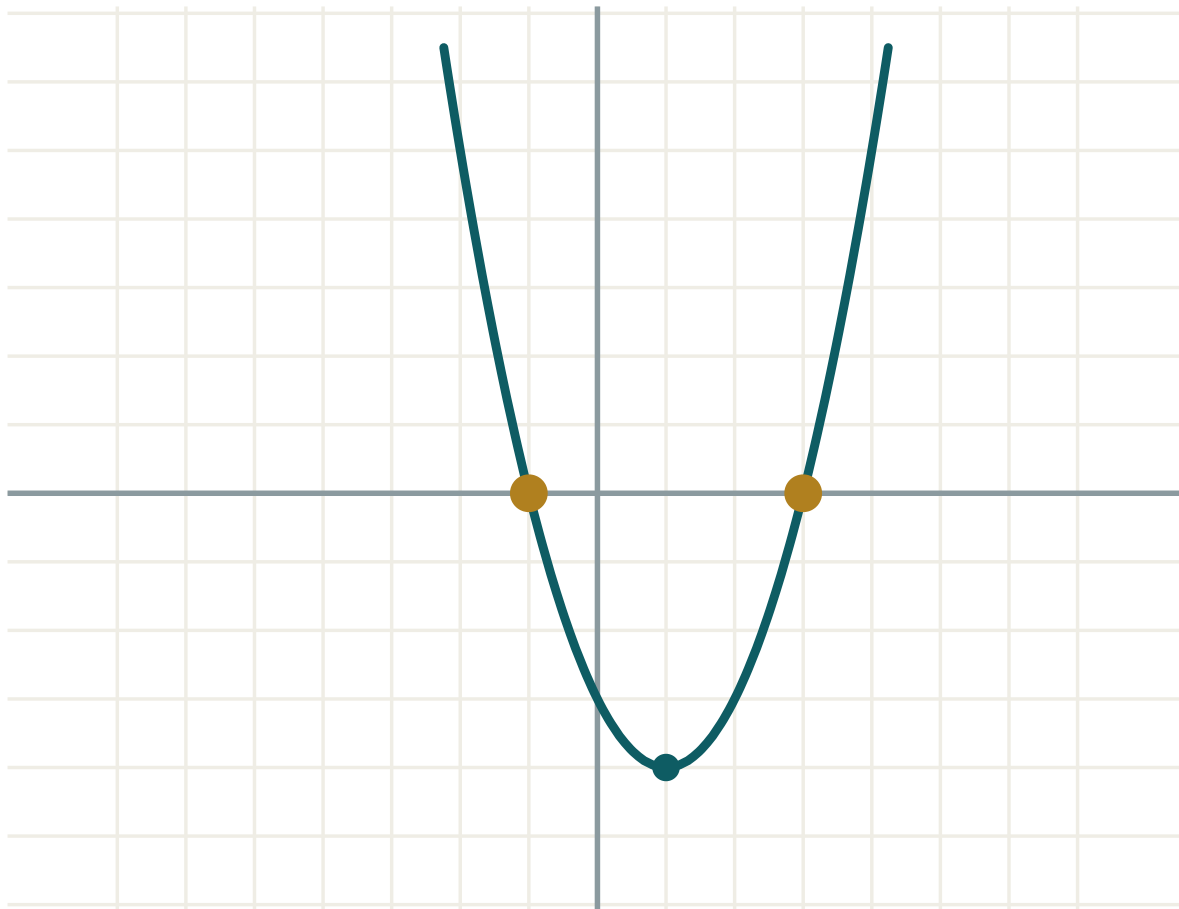
ANSWER

$(x - 3)^2 - 4$

03 Interactive model

Put the four skills on the board as four columns and let each pupil tick the ones they are sure of before the test, then compare with the result.

The discriminant in action

a **1**b **-2**c **-3** $D = b^2 - 4ac$ **16**roots **two** x_1 **-1** x_2 **3** $x_1 + x_2$ **2** $x_1 \cdot x_2$ **-3**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Quadratic equation	Kvadrat tenglama	Квадратное уравнение
Discriminant	Diskriminant	Дискриминант
Factorising	Ko'paytuvchilarga ajratish	Разложение на множители
Algebraic fraction	Algebraik kasr	Алгебраическая дробь
Arithmetic root	Arifmetik ildiz	Арифметический корень
Vieta's theorem	Viyet teoremasi	Теорема Виета
Diagnostic test	Tashxis testi	Диагностический тест
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$D = b^2 - 4ac$ is negative. The equation has:

two roots

one root

no real roots

infinitely many

For $x^2 + px + q = 0$, x_1x_2 equals:

p

$-p$

q

$-q$

$\sqrt{(-7)^2}$ equals:

-7

7

± 7

49

Cancelling $x + 3$ from $\frac{x^2 - 9}{x + 3}$ requires:

nothing

$x \neq -3$

$x \neq 3$

$x > 0$

$x^2 - 6x + 5$ completed is:

$(x - 3)^2 - 4$

$(x - 3)^2 + 4$

$(x - 6)^2 - 31$

$(x + 3)^2 - 4$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Factorise $x^2 - 7x + 12$

ANSWER $(x - 3)(x - 4)$

2. Factorise $x^2 - 25$

ANSWER $(x - 5)(x + 5)$

3. Solve $x^2 - 5x + 6 = 0$

ANSWER $x = 2, 3$

4. D of $2x^2 - 5x - 3$

ANSWER 49

5. Solve $2x^2 - 5x - 3 = 0$

ANSWER $x = 3, -0.5$

6. $\sqrt{50}$ in simplest form

ANSWER $5\sqrt{2}$

7. $\sqrt{(-7)^2}$

ANSWER 7

Medium · the standard question

7 PROBLEMS

1. Solve $x^2 + 4x + 7 = 0$

ANSWER No real roots — $D = -12$

2. Simplify $\frac{x^2 - 4}{x^2 + 4x + 4}$

ANSWER $\frac{x - 2}{x + 2}, x \neq -2$

3. $\sqrt{50} - \sqrt{18}$

ANSWER $2\sqrt{2}$

4. Complete the square: $x^2 - 6x + 5$

ANSWER $(x - 3)^2 - 4$

5. $x_1 + x_2$ and x_1x_2 for $x^2 - 8x + 15 = 0$

ANSWER 8 and 15

6. Solve $x^2 - 8x + 15 = 0$ by Vieta

ANSWER $x = 3, 5$

7. Factorise $2x^2 + 7x + 3$

ANSWER $(2x + 1)(x + 3)$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\frac{1}{x} + \frac{1}{x+1} = \frac{3}{2}$

ANSWER $x = 2$ or $x = -\frac{1}{3}$

2. For which k has $x^2 + kx + 9 = 0$ one repeated root?

ANSWER $k = \pm 6$

3. Simplify $\frac{x^2 - 9}{x^2 - 6x + 9} \cdot \frac{x - 3}{x + 3}$

ANSWER $1, x \neq \pm 3$

4. If x_1, x_2 are the roots of $x^2 - 5x + 3 = 0$, find $x_1^2 + x_2^2$

ANSWER 19

5. Solve $x^4 - 5x^2 + 4 = 0$

ANSWER $x = \pm 1, \pm 2$

6. Complete the square: $2x^2 - 8x + 3$

ANSWER $2(x - 2)^2 - 5$

7. For which m has $x^2 - 4x + m = 0$ no real roots?

ANSWER $m > 4$

07 Homework

Homework — 4 tasks

Bring the diagnostic back with the four skills ticked or crossed; the first chapter assumes all four.

1. Solve $3x^2 - 7x + 2 = 0$ by the formula and check the roots with Vieta's theorem.

2. Simplify $\frac{x^2 - 16}{x^2 - 8x + 16}$ and state the value of x that must be excluded.

3. Write $x^2 + 10x + 7$ in the form $(x + m)^2 + n$.

4. Name the one Grade 8 skill you are least sure of, and write in one sentence how you will practise it this week.

The quadratic function, and the graph of $y = x^2$

One curve, and every quadratic graph of the year is a moved and stretched copy of it.

LESSON 3–4

2 × 40 MIN

ALGEBRA 9, §§1–2

IGX 10.2

LESSON CLOCK

12'

22'

20'

20'

6'

What makes a function quadratic	12 min
Building the graph	22 min
Symmetry	20 min
Reading the graph	20 min
Homework	6 min

LEARNING OBJECTIVES

- State what makes a function quadratic.
- Draw $y = x^2$ from a table of values and name its parts.
- Use the symmetry of the parabola to halve the work.
- Read values and solve simple equations from the graph.

TEXTBOOK

National	pp. 5–9
Cambridge	Core & Extended, pp. 216–225
Workbook	Exercise 10.2

01

Explanation

What makes a function quadratic

$$y = ax^2 + bx + c, a \neq 0$$

The condition $a \neq 0$ is the whole definition. Lose the x^2 term and the function is linear; keep it and the graph is a parabola, whatever b and c do.

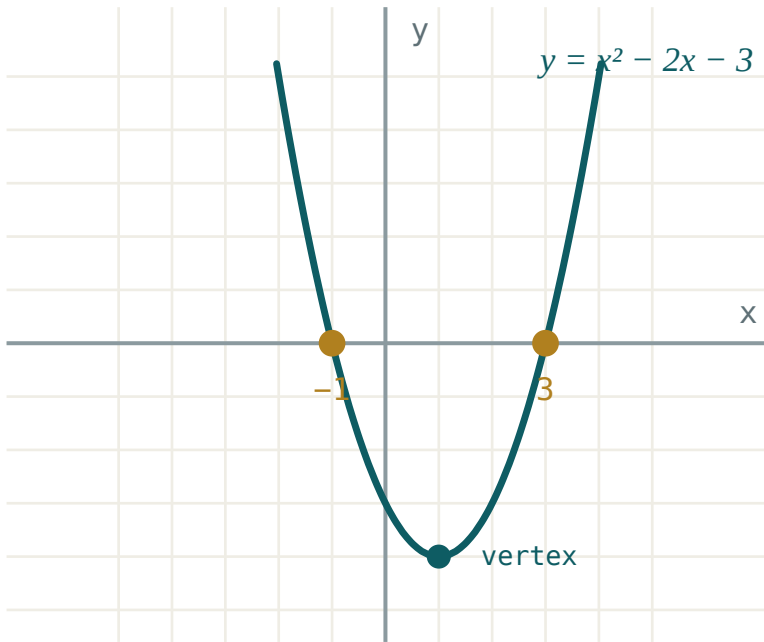
FUNCTION	QUADRATIC?	WHY
$y = x^2$	yes	$a = 1, b = c = 0$
$y = -3x^2 + \frac{1}{2}x$	yes	$a = -3$
$y = x^2 - 5x + 6$	yes	the usual form
$y = 2x + 7$	no	no x^2 term
$y = \frac{1}{x^2}$	no	x^2 is downstairs

WHERE THEY COME FROM

The area of a square of side x is x^2 . A stone thrown upwards at v m/s from height s_0 is at $s = -\frac{gt^2}{2} + vt + s_0$ after t seconds — a quadratic in t . The same curve describes a shape and a motion.

Building the graph

x	-3	-2	-1	0	1	2	3
$y = x^2$	9	4	1	0	1	4	9



$y = x^2$ — the vertex at the origin, the axis Oy , and both branches upward.

PART	WHERE
vertex	$(0, 0)$
axis of symmetry	$x = 0$, the Oy axis
branches	upward, because $a = 1 > 0$
least value	$y = 0$, at the vertex
range	$y \geq 0$

PLOT ENOUGH POINTS, AND JOIN WITH A CURVE

Three points are not a parabola — they are a guess. Take seven, and join them with one smooth sweep, not with straight segments. The vertex is a **turn**, never a corner.

Symmetry

$(-x)^2 = x^2$, so the two branches are mirror images in Oy . That halves the work: compute y for $x = 1, 2, 3$ and the negative side is free.

$f(-x) = f(x)$ — the parabola $y = x^2$ is an **even** function

TWO POINTS AT THE SAME HEIGHT

If $y = 9$ then $x = 3$ or $x = -3$: every horizontal line above the vertex meets the parabola **twice**, at points equidistant from the axis. Lesson 9 turns that single observation into the whole method for quadratic inequalities.

Reading the graph

QUESTION	READ OFF	ANSWER
the value of 1.5^2	up from $x = 1.5$	≈ 2.25
$\sqrt{5}$	across from $y = 5$	≈ 2.2
solve $x^2 = 4$	the line $y = 4$	$x = \pm 2$
solve $x^2 = -1$	the line $y = -1$	no solution

The last row is worth pausing on. The line $y = -1$ passes below the vertex and never meets the curve, so $x^2 = -1$ has no real solution — the same fact as $D < 0$, read from a picture instead of a formula.

02 Worked examples

EXAMPLE 1 Which of $y = 5x^2$, $y = 5x$, $y = 5 - x^2$ are quadratic?

- ① $y = 5x^2$: $a = 5 \neq 0$ — yes.
- ② $y = 5x$: no x^2 term — no.
- ③ $y = 5 - x^2$: $a = -1 \neq 0$ — yes.

ANSWER

The first and the third

EXAMPLE 2 Use the graph of $y = x^2$ to solve $x^2 = 6.25$.

- ① Draw the line $y = 6.25$.
- ② It meets the curve twice, symmetrically.
- ③ $x = \pm 2.5$
Check: $2.5^2 = 6.25$.

ANSWER

$x = 2.5$ and $x = -2.5$

EXAMPLE 3 The point $(a, 49)$ lies on $y = x^2$. Find a .

① $a^2 = 49$

② $a = \pm 7$

Two points, by symmetry.

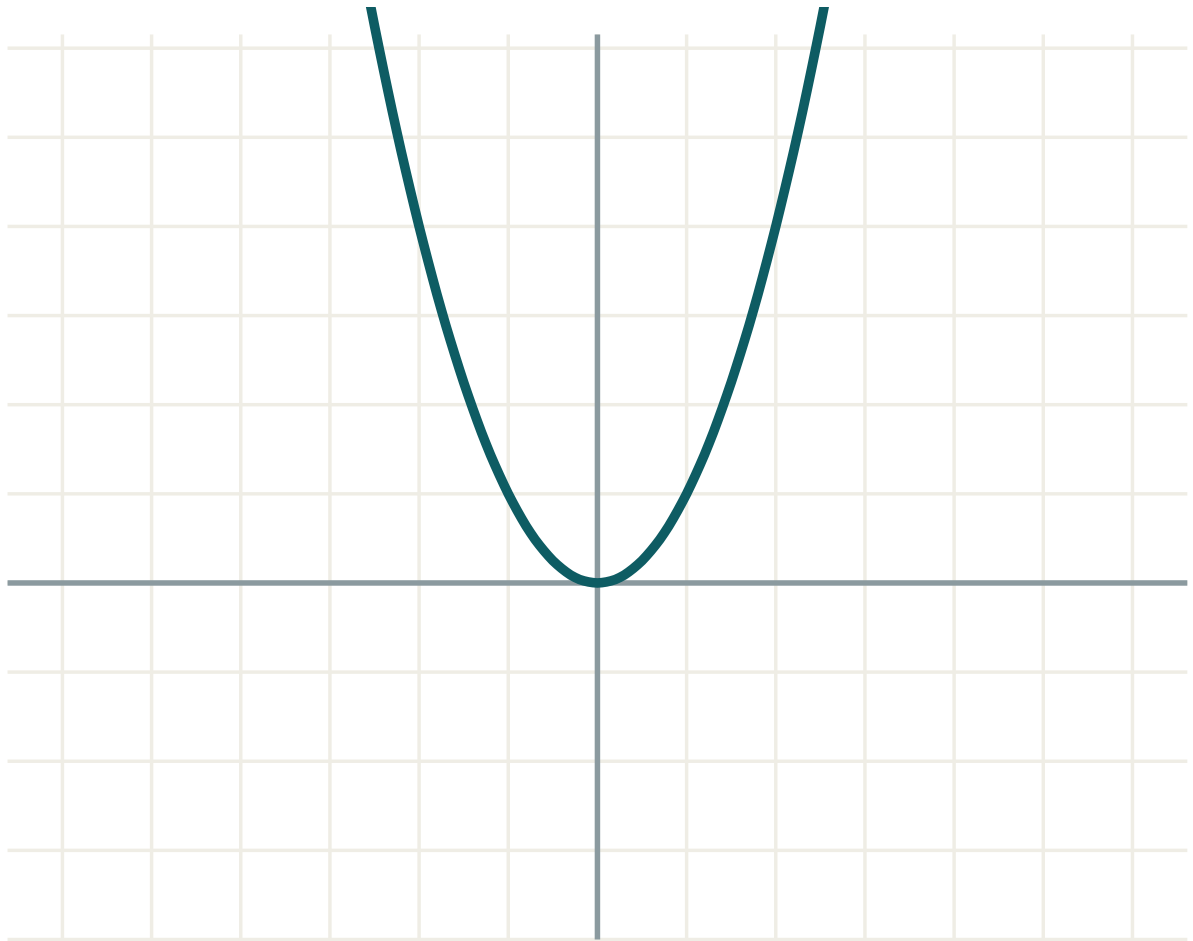
ANSWER

$a = 7$ or $a = -7$

03 Interactive model

Hang a light chain from two points — the class sees a curve that looks parabolic, then checks whether it really is by measuring.

The parabola, moved and stretched



a 1

b 0

c 0

equation $y = f(x)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Quadratic function	Kvadratik funksiya	Квадратичная функция
Parabola	Parabola	Парабола
Vertex	Uchi	Вершина
Axis of symmetry	Simmetriya o'qi	Ось симметрии
Branch	Tarmoq	Ветвь
Leading coefficient	Bosh koeffitsiyent	Старший коэффициент
Table of values	Qiymatlar jadvali	Таблица значений
Argument	Argument	Аргумент

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$y = ax^2 + bx + c$ is quadratic when:

$a \neq 0$

$b \neq 0$

$c \neq 0$

always

The vertex of $y = x^2$ is:

$(1, 1)$

$(0, 0)$

$(0, 1)$

$(-1, 1)$

Its axis of symmetry is:

$y = 0$

$x = 0$

$y = x$

none

$x^2 = 9$ has:

one solution

two solutions

no solution

three

$x^2 = -4$ has:

one solution

two solutions

no real solution

$x = -2$

$y = x^2$ is an even function because:

x^2 is positive

$f(-x) = f(x)$

it is a curve

of the vertex

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. y at $x = 4$ on $y = x^2$

ANSWER 16

2. y at $x = -4$

ANSWER 16

3. Vertex of $y = x^2$

ANSWER $(0, 0)$

4. Axis of symmetry of $y = x^2$

ANSWER $x = 0$

5. Solve $x^2 = 25$

ANSWER $x = \pm 5$

6. Is $y = 3x^2$ quadratic?

ANSWER Yes

7. Is $y = 3x + 1$ quadratic?

ANSWER No

Medium · the standard question

7 PROBLEMS

1. Solve $x^2 = 6.25$

ANSWER $x = \pm 2.5$

2. Solve $x^2 = -4$

ANSWER No real solution

3. a if $(a, 49)$ lies on $y = x^2$

ANSWER $a = \pm 7$

4. Range of $y = x^2$

ANSWER $y \geq 0$

5. Least value of $y = x^2$

ANSWER 0, at $x = 0$

6. Which is quadratic: $y = 5x^2$, $y = 5x$, $y = 5 - x^2$?

ANSWER The first and the third

7. Estimate $\sqrt{5}$ from the graph

ANSWER ≈ 2.2

Hard · stretch and reason

7 PROBLEMS

1. For which c does $y = x^2$ meet $y = c$ exactly once?

ANSWER $c = 0$

2. The chord of $y = x^2$ at height $h > 0$: its length

ANSWER $2\sqrt{h}$

3. Points of $y = x^2$ at distance 5 from the origin

ANSWER $(\pm 2, 4)$

4. Where does $y = x^2$ meet $y = x$?

ANSWER $(0, 0)$ and $(1, 1)$

5. Where does $y = x^2$ meet $y = 2x + 3$?

ANSWER $(-1, 1)$ and $(3, 9)$

6. Show that $y = x^2$ lies above $y = x$ for $x > 1$

ANSWER $x^2 - x = x(x - 1) > 0$

7. A square of side x has area 30: estimate x from the graph

ANSWER ≈ 5.5

07 Homework

Homework — 4 tasks

Draw the graph on squared paper with seven points; keep it — the next four lessons all use it.

1. Make a table of $y = x^2$ for x from -4 to 4 and draw the graph.
2. From your graph, estimate 2.5^2 , $\sqrt{7}$ and the solutions of $x^2 = 10$.
3. Find both points of $y = x^2$ whose y -coordinate is 64.
4. Explain in two sentences why $x^2 = -9$ has no solution, using the graph rather than the discriminant.

The graphs of $y = ax^2$ and $y = ax^2 + bx + c$

a decides the shape, and completing the square moves the whole curve without changing it.

LESSON 5–6

2 × 40 MIN

ALGEBRA 9, §§3–4

IGX 10.2

LESSON CLOCK

14'

18'

20'

16'

12'

What a does	14 min
Moving the parabola	18 min
Completing the square	20 min
Reading either form	16 min
Homework	12 min

LEARNING OBJECTIVES

- Describe the effect of a on the shape and direction of the parabola.
- Recognise $y = a(x - m)^2 + n$ as $y = ax^2$ translated.
- Convert $y = ax^2 + bx + c$ to vertex form by completing the square.
- Read the vertex, the axis and the least or greatest value from either form.

TEXTBOOK

National	pp. 10–17
Cambridge	Core & Extended, pp. 216–225
Workbook	Exercise 10.2

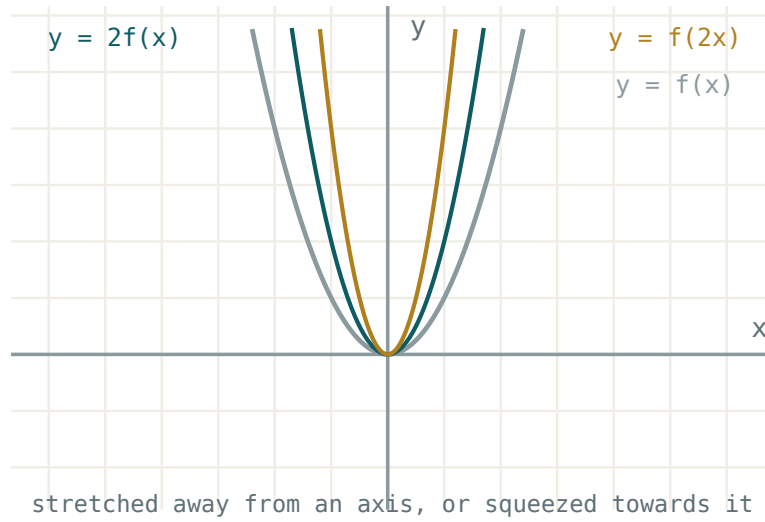
01

Explanation

What a does

$y = ax^2$ is $y = x^2$ with every height multiplied by a .

A	BRANCHES	SHAPE
$a > 1$	upward	narrower than $y = x^2$
$0 < a < 1$	upward	wider
$a < 0$	downward	reflected in Ox



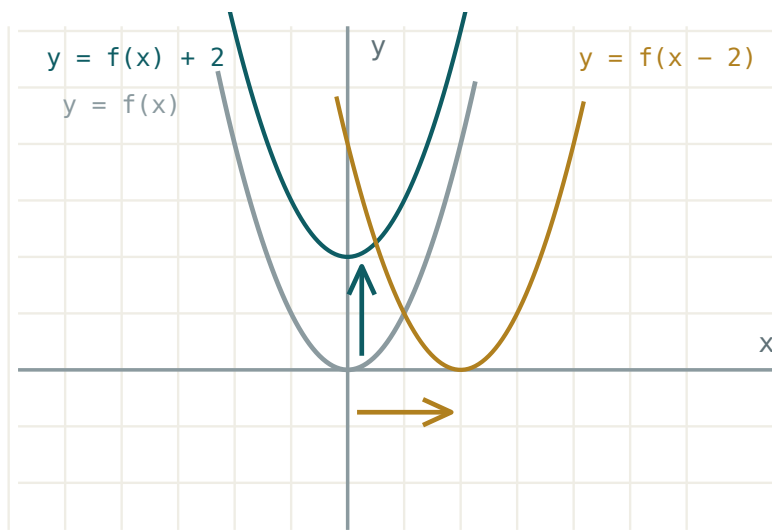
The same parabola stretched — a changes the height of every point.

THE SIGN OF a ANSWERS HALF THE QUESTION

$a > 0$: the vertex is the **lowest** point, and the function has a least value and no greatest. $a < 0$: the vertex is the **highest** point, and it is the other way round. Decide the sign before anything else and the rest follows.

Moving the parabola

$y = a(x - m)^2 + n$ — the vertex is at (m, n)



Along by m , up by n — the curve itself never changes.

EQUATION	VERTEX	AXIS	LEAST OR GREATEST
$y = x^2$	$(0, 0)$	$x = 0$	least 0
$y = (x - 3)^2$	$(3, 0)$	$x = 3$	least 0
$y = (x - 3)^2 - 4$	$(3, -4)$	$x = 3$	least -4
$y = -2(x + 1)^2 + 5$	$(-1, 5)$	$x = -1$	greatest 5

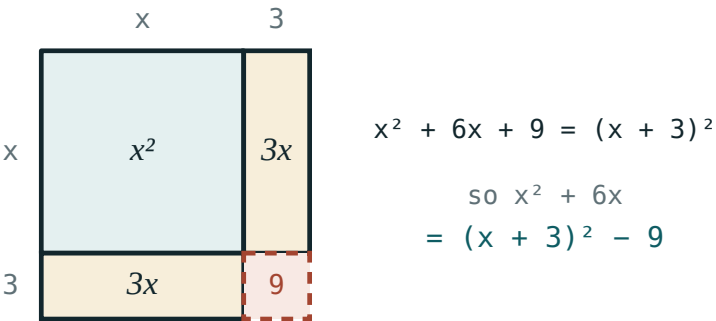
THE BRACKET SIGN IS REVERSED

$(x - 3)^2$ moves the curve **right** by 3; $(x + 1)^2$ moves it **left** by 1. Read $(x + 1)$ as $(x - (-1))$ and the trap disappears.

Completing the square

Any $y = ax^2 + bx + c$ becomes $y = a(x - m)^2 + n$ in three lines.

STEP	$Y = 2X^2 - 12X + 13$
take out a	$2(x^2 - 6x) + 13$
complete inside	$2[(x - 3)^2 - 9] + 13$
expand and tidy	$2(x - 3)^2 - 18 + 13 = 2(x - 3)^2 - 5$



The square completed — the constant that had to be added, and taken away again.

OR USE THE FORMULA FOR THE AXIS

$$x_{vertex} = -\frac{b}{2a}, \text{ then } y_{vertex} = f(x_{vertex})$$

For $2x^2 - 12x + 13$: $x = \frac{12}{4} = 3, y = f(3) = -5$ — the same vertex $(3, -5)$ in one line. Completing the square is worth learning anyway, because it is how the quadratic formula itself is proved.

Reading either form

WANTED	FROM $A(X - M)^2 + N$	FROM $AX^2 + BX + C$
vertex	(m, n)	$(-\frac{b}{2a}, f(-\frac{b}{2a}))$
axis	$x = m$	$x = -\frac{b}{2a}$
y-intercept	$am^2 + n$	c
roots	$m \pm \sqrt{-\frac{n}{a}}$	$\frac{-b \pm \sqrt{D}}{2a}$

Each form answers one question instantly and the other with a little work. The vertex form gives the turning point; the expanded form gives the y-intercept. Convert whichever way the question points.

02 Worked examples

EXAMPLE 1 Write $y = 2x^2 - 12x + 13$ in vertex form and state the least value.

① $2(x^2 - 6x) + 13$

② $2[(x - 3)^2 - 9] + 13$

③ $2(x - 3)^2 - 5$

④ Least value -5 at $x = 3$.
 $a > 0$, so the vertex is lowest.

ANSWER

$y = 2(x - 3)^2 - 5$; least value -5

EXAMPLE 2 Find the vertex of $y = -x^2 + 6x - 5$ and say whether it is a maximum.

① $x = -\frac{6}{-2} = 3$

② $y = -9 + 18 - 5 = 4$

③ $a = -1 < 0$

④ Vertex $(3, 4)$ — a maximum.

ANSWER

$(3, 4)$, a greatest value of 4

EXAMPLE 3 A parabola has vertex $(-2, 1)$ and passes through $(0, 9)$. Find its equation.

① $y = a(x + 2)^2 + 1$

② $9 = a(4) + 1$
Substitute $(0, 9)$.

③ $a = 2$

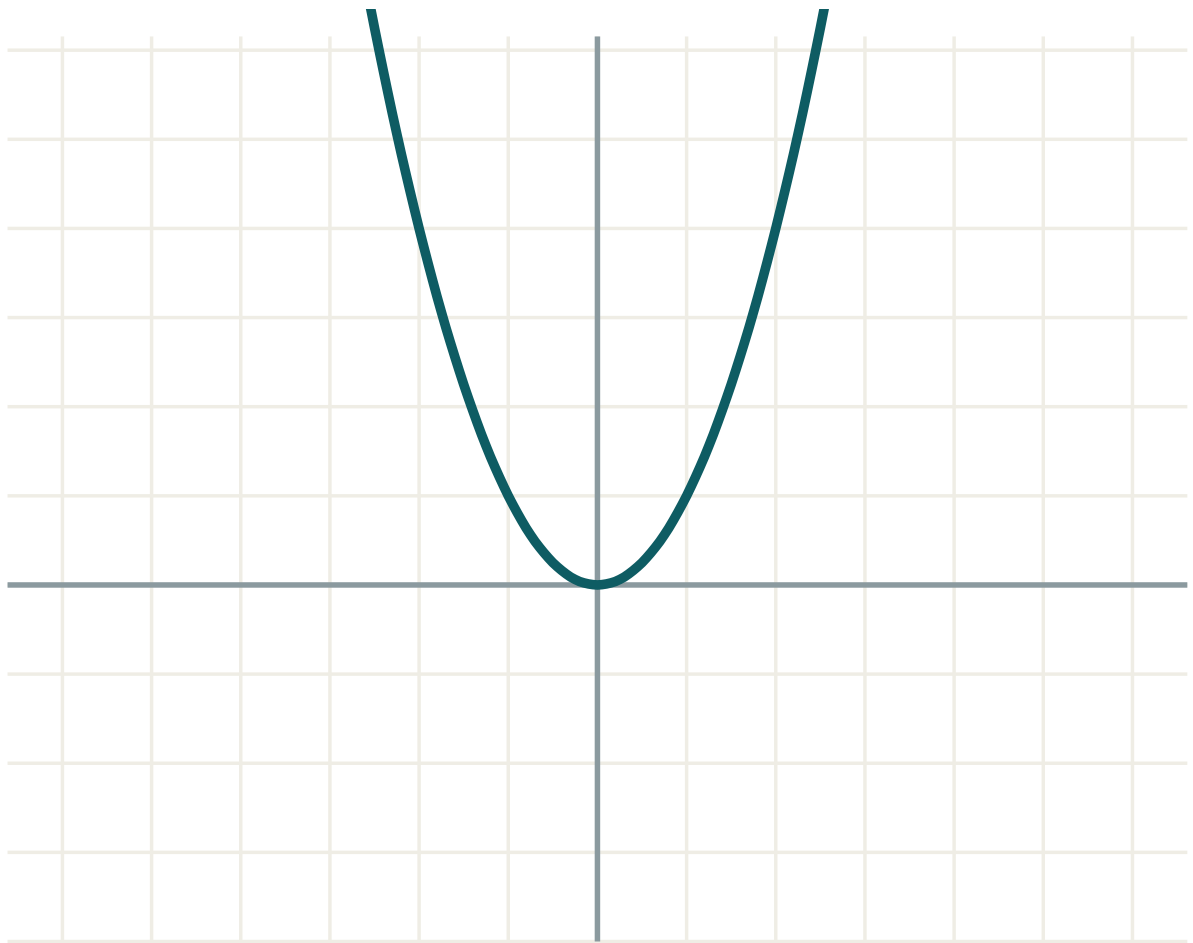
④ $y = 2(x + 2)^2 + 1$

ANSWER

$$y = 2(x + 2)^2 + 1$$

03 Interactive model

Draw $y = x^2$ once on the board and then slide a tracing-paper copy of it — every equation of the lesson is that one curve in a new place.

a, m and n on one parabola**a 1****b 0****c 0****equation $y = f(x)$** **04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Leading coefficient	Bosh koeffitsiyent	Старший коэффициент
Stretch	Cho'zish	Растяжение
Translation	Parallel ko'chirish	Параллельный перенос
Vertex form	Uchi ko'rinishi	Вершинная форма
Completing the square	To'la kvadratga keltirish	Выделение полного квадрата
Greatest value	Eng katta qiymat	Наибольшее значение
Least value	Eng kichik qiymat	Наименьшее значение
Axis of symmetry	Simmetriya o'qi	Ось симметрии

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$a < 0$ means the branches point:

up

down

sideways

either way

The vertex of $y = (x - 4)^2 + 3$ is:

$(4, 3)$

$(-4, 3)$

$(4, -3)$

$(3, 4)$

The vertex of $y = (x + 1)^2 - 2$ is:

$(1, -2)$

$(-1, -2)$

$(-1, 2)$

$(1, 2)$

The axis of $y = ax^2 + bx + c$ is:

$$x = \frac{b}{2a}$$

$$x = -\frac{b}{2a}$$

$$x = \frac{b}{a}$$

$$x = c$$

For $a > 0$ the vertex gives:

a greatest value

a least value

neither

both

$y = 2(x - 3)^2 - 5$ has y -intercept:

-5

13

3

2

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Vertex of $y = (x - 2)^2$

ANSWER $(2, 0)$

2. Vertex of $y = x^2 + 5$

ANSWER $(0, 5)$

3. Vertex of $y = (x + 3)^2 - 1$

ANSWER $(-3, -1)$

4. Branches of $y = -4x^2$

ANSWER Downward

5. Which is narrower, $y = 3x^2$ or $y = \frac{1}{3}x^2$?

ANSWER $y = 3x^2$

6. Axis of $y = x^2 - 6x + 1$

ANSWER $x = 3$

7. y-intercept of $y = 2x^2 - 5x + 7$

ANSWER 7

Medium · the standard question

7 PROBLEMS

1. Vertex form of $y = x^2 - 6x + 5$

ANSWER $(x - 3)^2 - 4$

2. Vertex form of $y = 2x^2 - 12x + 13$

ANSWER $2(x - 3)^2 - 5$

3. Vertex of $y = -x^2 + 6x - 5$

ANSWER $(3, 4)$, a maximum

4. Least value of $y = x^2 + 4x + 9$

ANSWER 5 at $x = -2$

5. Greatest value of $y = -2x^2 + 8x - 3$

ANSWER 5 at $x = 2$

6. Equation with vertex $(-2, 1)$ through $(0, 9)$

ANSWER $y = 2(x + 2)^2 + 1$

7. Vertex form of $y = 3x^2 + 6x$

ANSWER $3(x + 1)^2 - 3$

Hard · stretch and reason

7 PROBLEMS

1. For which k is the least value of $y = x^2 - 4x + k$ equal to 3?

ANSWER $k = 7$

2. A parabola through $(0, 3)$, $(1, 0)$, $(4, 3)$: find its equation

ANSWER $y = x^2 - 4x + 3$

3. The vertex of $y = x^2 + bx + 7$ lies on Ox : find b

ANSWER $b = \pm 2\sqrt{7}$

4. Range of $y = -3(x - 1)^2 + 12$

ANSWER $y \leq 12$

5. Show that $y = x^2 + px + q$ has least value $q - \frac{p^2}{4}$

ANSWER Complete the square

6. A ball follows $h = -5t^2 + 20t$: its greatest height and when

ANSWER 20 m at $t = 2$ s

7. Two parabolas $y = x^2$ and $y = -x^2 + 8$: where do they meet?

ANSWER $(\pm 2, 4)$

07 Homework

Homework — 5 tasks

Write the vertex on its own line before drawing anything.

- Write $y = x^2 - 8x + 11$ in vertex form and state the vertex and the least value.
- Write $y = -2x^2 + 4x + 1$ in vertex form and state the greatest value.
- Find the equation of the parabola with vertex $(1, -3)$ passing through $(3, 5)$.
- Sketch $y = (x - 2)^2 - 9$, marking the vertex, the axis and both roots.

5. Explain in two sentences why $(x + 4)^2$ moves the graph left and not right.

Drawing the graph of a quadratic function

Five marks on the paper — vertex, axis, two roots and the intercept — and the curve draws itself.

LESSON 7–8

2 × 40 MIN

ALGEBRA 9, §5

IGX 10.2

LESSON CLOCK

13'

22'

18'

18'

9'

The five marks 13 min

A worked sketch 22 min

Reading the sketch 18 min

Matching graphs 18 min

Homework 9 min

LEARNING OBJECTIVES

- Sketch a parabola from its vertex, roots and y-intercept.
- Use the discriminant to decide whether the roots exist before looking for them.
- Read the range, and the intervals where the function is positive or negative.
- Match an equation to a graph and a graph to an equation.

TEXTBOOK

National pp. 18–23

Cambridge Core & Extended,
pp. 216–225

Workbook Exercise 10.2

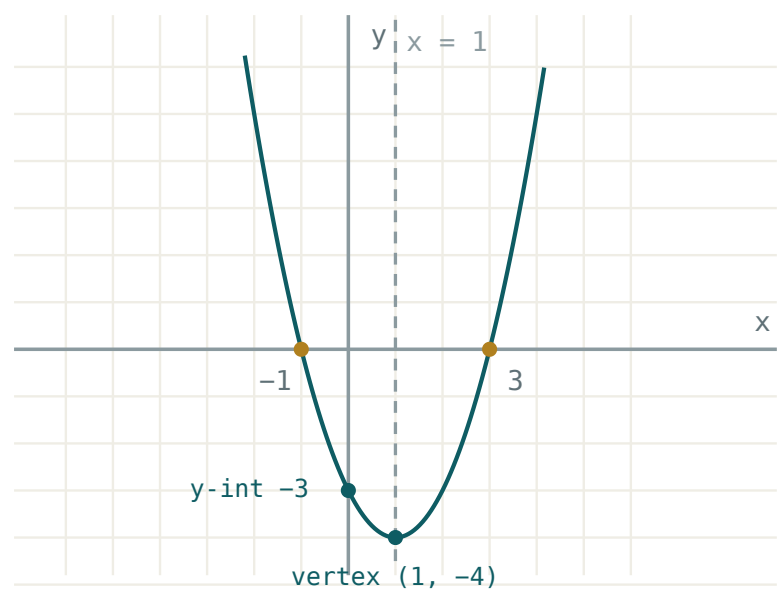
01 Explanation

The five marks

EVERY SKETCH, IN THE SAME ORDER

1. The **sign of a** — up or down. Decide first, draw last.
2. The **y-intercept**: put $x = 0$, giving c . Free.
3. The **roots**: solve $ax^2 + bx + c = 0$, if $D \geq 0$.
4. The **axis**: $x = -\frac{b}{2a}$, or halfway between the roots.
5. The **vertex**: on the axis, at $y = f(-\frac{b}{2a})$.

Mark all five, then join them with one smooth curve.



The five marks, and the curve through them.

A worked sketch

Sketch $y = x^2 - 2x - 3$.

STEP	WORKING	MARK
direction	$a = 1 > 0$	opens upward
y-intercept	$c = -3$	$(0, -3)$
roots	$(x - 3)(x + 1) = 0$	$(3, 0), (-1, 0)$
axis	$x = \frac{3 + (-1)}{2} = 1$	$x = 1$
vertex	$f(1) = 1 - 2 - 3 = -4$	$(1, -4)$

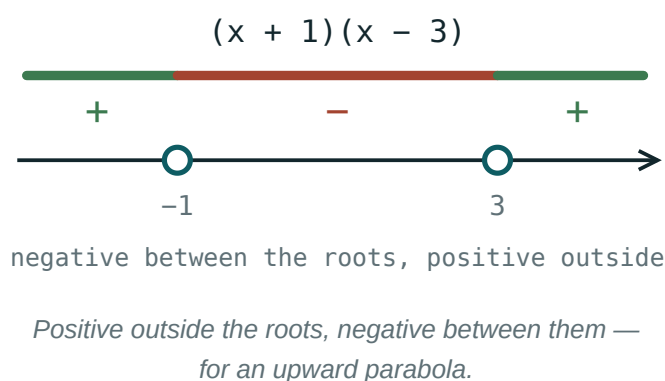
THE AXIS IS HALFWAY BETWEEN THE ROOTS, ALWAYS

$\frac{x_1 + x_2}{2}$ is the same number as $-\frac{b}{2a}$ — Vieta guarantees it. If your two answers disagree, one of them is wrong, and the check costs five seconds.

Reading the sketch

Once the picture exists, four more questions are free.

QUESTION	FOR $Y = X^2 - 2X - 3$
range	$y \geq -4$
where $y > 0$	$x < -1$ or $x > 3$
where $y < 0$	$-1 < x < 3$
decreasing on	$x \leq 1$
increasing on	$x \geq 1$



OUTSIDE AND BETWEEN

For $a > 0$: **positive outside** the roots, **negative between** them. For $a < 0$ it is the other way round. Lessons 9–12 are this one sentence, applied.

Matching graphs

Given four graphs and four equations, three checks decide every pairing.

LOOK AT	IT TELLS YOU
the direction of the branches	the sign of a
where the curve cuts Oy	c
how many times it cuts Ox	the sign of D

Example. A downward parabola cutting Oy at 4 and missing Ox needs $a < 0$, $c = 4$ and $D < 0$. Among $y = -x^2 + 4$, $y = -x^2 + x - 4$, $y = -x^2 - x + 4$ and $y = x^2 + 4$, only the second has $c = -4$, so it is out on the intercept; the first and third cut Ox ; so it is none of them — and a fourth equation is needed. Checking all three, and saying when none fits, is the skill.

02 Worked examples

EXAMPLE 1 Sketch $y = x^2 - 2x - 3$, marking all five features.

- ① $a = 1 > 0$ — upward.
- ② $(0, -3)$; roots $x = -1, 3$.
- ③ Axis $x = 1$.
Halfway between the roots.
- ④ Vertex $(1, -4)$.

ANSWER

Vertex $(1, -4)$, roots -1 and 3

EXAMPLE 2 Sketch $y = -x^2 + 4x$ and state its range.

- ① $a = -1 < 0$ — downward.
- ② $-x(x - 4) = 0$, so roots 0 and 4 .
- ③ Axis $x = 2$; vertex $(2, 4)$.
- ④ Range $y \leq 4$.
The vertex is highest.

ANSWER

Vertex $(2, 4)$; range $y \leq 4$

EXAMPLE 3 A parabola opens upward, cuts Oy at 6 and touches Ox . What can be said about it?

- ① $a > 0$ and $c = 6$.
- ② Touching means $D = 0$, so $b^2 = 4ac = 24a$.
- ③ With $a = 1$: $b = \pm 2\sqrt{6}$.
- ④ The repeated root is $-\frac{b}{2} = \mp\sqrt{6}$.

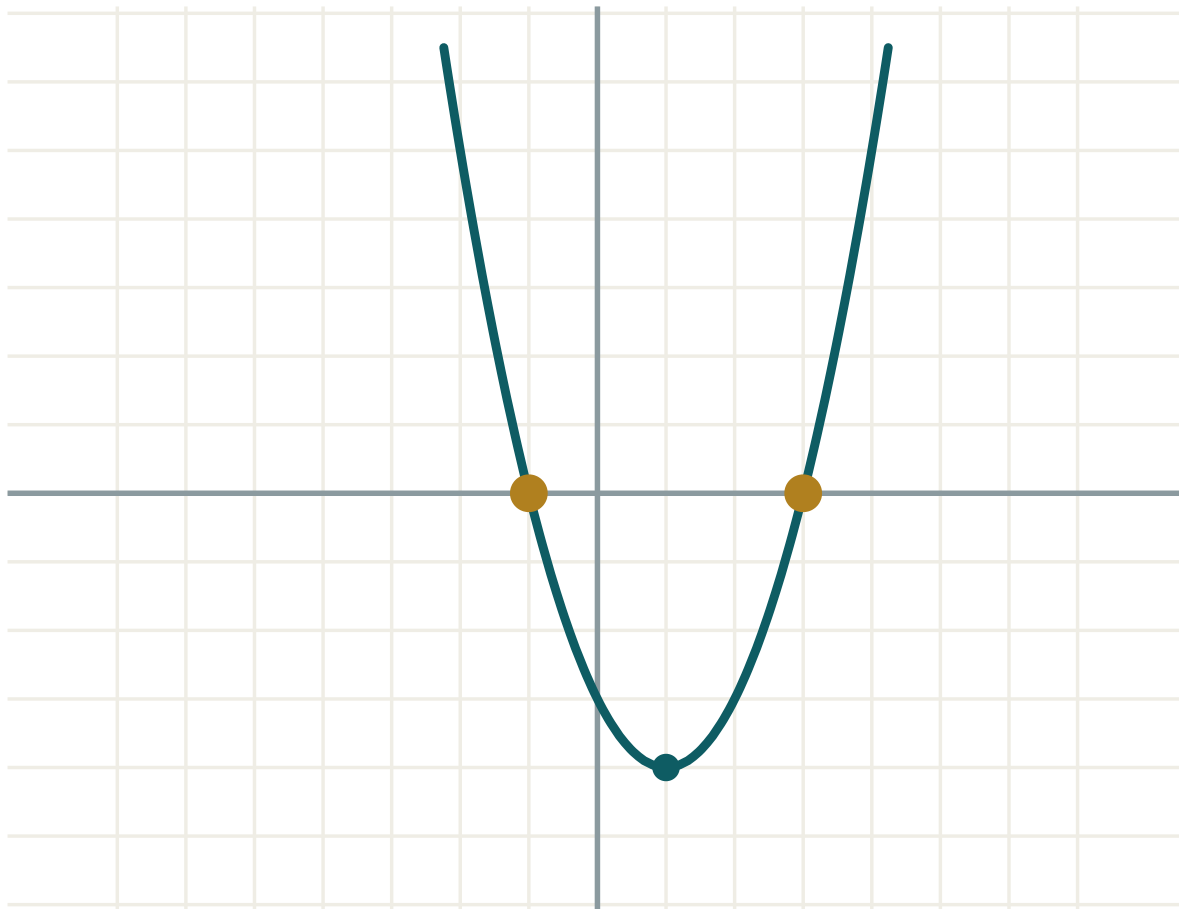
ANSWER

$y = x^2 \pm 2\sqrt{6}x + 6$ is one such parabola

03 Interactive model

Give each pupil one equation and one blank grid; pin the finished sketches side by side and let the class find the two that disagree.

Five marks, one curve



a 1

b -2

c -3

 $D = b^2 - 4ac$ 16

roots two

 x_1 -1 x_2 3 $x_1 + x_2$ 2 $x_1 \cdot x_2$ -3

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Sketch	Eskiz	Эскиз
Root (zero)	Nol	Ноль функции
Intercept	Kesish nuqtasi	Точка пересечения
Range	Qiymatlar sohasi	Область значений
Discriminant	Diskriminant	Дискриминант
Positive interval	Musbat oraliq	Промежуток положительности
Symmetry	Simmetriya	Симметрия
Turning point	Burilish nuqtasi	Точка поворота

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The axis of symmetry lies:

at the y -intercept

halfway between the roots

at $x = 0$

at the larger root

For $a > 0, y > 0$:

between the roots

outside the roots

everywhere

nowhere

The y -intercept of $y = ax^2 + bx + c$ is:

a

b

c

0

A parabola touching Ox has:

$$D > 0$$

$$D = 0$$

$$D < 0$$

$$a = 0$$

The range of $y = -x^2 + 4x$ is:

$$y \geq 4$$

$$y \leq 4$$

$$y \geq 0$$

all y

$y = x^2 - 2x - 3$ decreases on:

$$x \leq 1$$

$$x \geq 1$$

all x

$$x \leq 0$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Roots of $y = x^2 - 2x - 3$

ANSWER -1 and 3

2. Axis of the same

ANSWER $x = 1$

3. Vertex of the same

ANSWER $(1, -4)$

4. y -intercept of the same

ANSWER -3

5. Direction of $y = -2x^2 + x$

ANSWER Downward

6. Roots of $y = x^2 - 9$

ANSWER ± 3

7. Vertex of $y = x^2 - 9$

ANSWER $(0, -9)$

Medium · the standard question

7 PROBLEMS

1. Sketch data for $y = -x^2 + 4x$

ANSWER Roots 0, 4; vertex (2, 4)

2. Range of $y = x^2 - 2x - 3$

ANSWER $y \geq -4$

3. Where is $x^2 - 2x - 3 > 0$?

ANSWER $x < -1$ or $x > 3$

4. Where is $x^2 - 2x - 3 < 0$?

ANSWER $-1 < x < 3$

5. Sketch data for $y = x^2 - 6x + 9$

ANSWER Touches Ox at (3, 0)

6. Sketch data for $y = x^2 + 2x + 5$

ANSWER No roots; vertex (-1, 4)

7. Increasing interval of $y = -x^2 + 4x$

ANSWER $x \leq 2$

Hard · stretch and reason

7 PROBLEMS

1. A parabola with roots -2 and 5 through $(0, -10)$

ANSWER $y = x^2 - 3x - 10$

2. A parabola with vertex $(2, -1)$ and y -intercept 3

ANSWER $y = (x - 2)^2 - 1$

3. For which k does $y = x^2 + kx + 4$ touch Ox ?

ANSWER $k = \pm 4$

4. For which k does $y = x^2 + kx + 4$ miss Ox ?

ANSWER $-4 < k < 4$

5. Where is $-x^2 + 4x > 0$?

ANSWER $0 < x < 4$

6. A parabola opens upward, touches Ox and cuts Oy at 6 : give one equation

ANSWER $y = x^2 - 2\sqrt{6}x + 6$

7. Two parabolas $y = x^2 - 4$ and $y = 4 - x^2$: the area they enclose is bounded by which roots?

ANSWER $x = \pm 2$

07 Homework

Homework — 5 tasks

Mark the five features on every sketch, with their coordinates written beside them.

1. Sketch $y = x^2 - 4x + 3$, marking the vertex, the axis, both roots and the intercept.
2. Sketch $y = -x^2 - 2x + 3$ and state its range.
3. Sketch $y = x^2 + 2x + 3$ and explain why it has no roots.
4. Find the equation of the parabola with roots -3 and 1 passing through $(0, -6)$.

5. For $y = x^2 - 4x + 3$, write down the intervals where the function is positive and where it is negative.

The quadratic inequality and its solution

Solve the equation, then decide which side of the roots the question wants.

LESSON 9–10

2 × 40 MIN

ALGEBRA 9, §6

EXTENSION BEYOND IGX

LESSON CLOCK

11'

22'

22'

18'

7'

Standard form first	11 min
Outside and between	22 min
The three cases	22 min
Writing the answer	18 min
Homework	7 min

LEARNING OBJECTIVES

- Recognise a quadratic inequality and write it in standard form.
- Find the roots, then choose the correct interval.
- Handle the three cases $D > 0$, $D = 0$ and $D < 0$.
- Write the answer as an interval or a union of intervals.

TEXTBOOK

National	pp. 24–27
Cambridge	Core & Extended, pp. 216–225
Workbook	Exercise 10.2

01

Explanation

Standard form first

Every quadratic inequality is first pushed to one side, so that it compares a trinomial with zero:

$$ax^2 + bx + c > 0, < 0, \geq 0 \text{ or } \leq 0$$

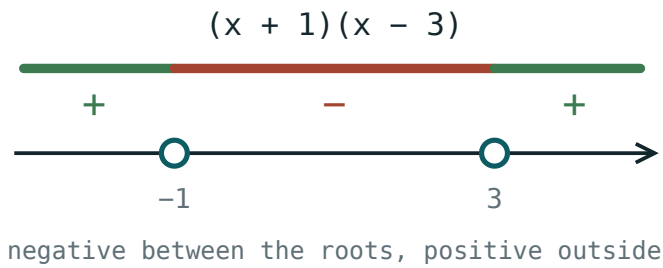
AS WRITTEN	IN STANDARD FORM
$x^2 > 3x + 4$	$x^2 - 3x - 4 > 0$
$2x^2 \leq 5x$	$2x^2 - 5x \leq 0$
$(x - 1)(x + 3) < 5$	$x^2 + 2x - 8 < 0$

NEVER DIVIDE AN INEQUALITY BY x

From $2x^2 \leq 5x$ you may **not** divide by x to get $2x \leq 5$ — the sign of x is unknown, and $x = 0$ is lost. Move everything to one side and factorise instead: $x(2x - 5) \leq 0$.

Outside and between

Once the roots $x_1 < x_2$ are known, the picture decides everything.



*An upward parabola: positive outside the roots,
negative between them.*

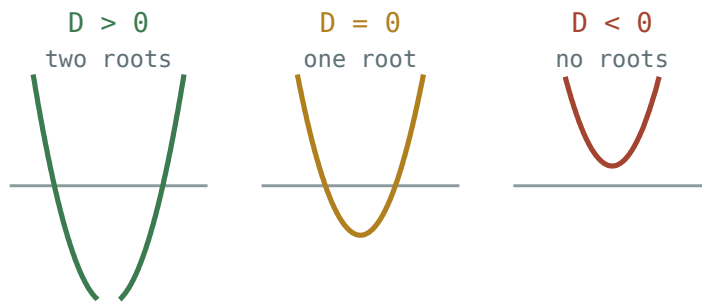
INEQUALITY	$A > 0$	$A < 0$
$ax^2 + bx + c > 0$	$x < x_1$ or $x > x_2$	$x_1 < x < x_2$
$ax^2 + bx + c < 0$	$x_1 < x < x_2$	$x < x_1$ or $x > x_2$

ONE SENTENCE FOR ALL FOUR ROWS

The trinomial takes the **sign of a** outside the roots and the **opposite sign** between them. Sketch the parabola, mark the roots, and read the answer off — no table needed.

The three cases

D	PICTURE ($A > 0$)	> 0	< 0
$D > 0$	cuts Ox twice	outside	between
$D = 0$	touches at x_0	all $x \neq x_0$	no solution
$D < 0$	entirely above Ox	all x	no solution

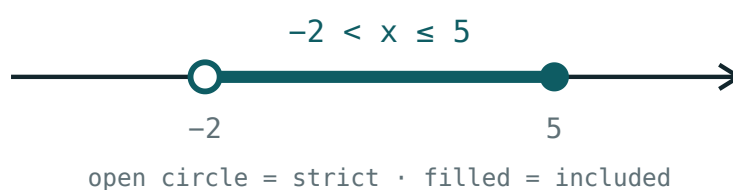


The three positions, and the answers that follow from each.

The two degenerate cases are the ones examiners like. $x^2 + 2x + 5 > 0$ has $D = -16 < 0$ and $a > 0$, so the parabola never touches the axis: the answer is **every** real x . And $x^2 + 2x + 5 < 0$ has **no** solution at all. Neither needs a single root to be found.

Writing the answer

IN WORDS	IN INTERVALS	ON THE LINE
between -1 and 4	$(-1, 4)$	open circles
between, inclusive	$[-1, 4]$	filled circles
outside	$(-\infty, -1) \cup (4, +\infty)$	two rays
everything but 3	$(-\infty, 3) \cup (3, +\infty)$	one hole



The interval drawn — open circles for a strict inequality, filled for a weak one.

∞ ALWAYS TAKES A ROUND BRACKET

$[4, +\infty)$, never $[4, +\infty]$. Infinity is not a number the variable can equal, so the bracket beside it is round whatever the inequality sign is.

02 Worked examples

EXAMPLE 1 Solve $x^2 - 3x - 4 > 0$.

① $x^2 - 3x - 4 = 0 \Rightarrow (x - 4)(x + 1) = 0$

② $x_1 = -1, x_2 = 4.$

③ $a = 1 > 0$ and we want > 0 — outside.

④ $x < -1$ or $x > 4.$

ANSWER

$(-\infty, -1) \cup (4, +\infty)$

EXAMPLE 2 Solve $2x^2 \leq 5x$.

① $2x^2 - 5x \leq 0$
Standard form.

② $x(2x - 5) \leq 0$
Factor — do not divide.

③ Roots 0 and 2.5; $a > 0$, want ≤ 0 — between.

④ $0 \leq x \leq 2.5$

ANSWER

$[0, 2.5]$

EXAMPLE 3 Solve $x^2 + 2x + 5 > 0$.

- ① $D = 4 - 20 = -16 < 0$
- ② $a = 1 > 0$, so the parabola lies entirely above Ox .
- ③ The trinomial is positive for every x .

ANSWERAll real x **EXAMPLE 4** Solve $-x^2 + 4x - 3 \geq 0$.

- ① $x^2 - 4x + 3 \leq 0$
Multiply by -1 and flip the sign.
- ② $(x - 1)(x - 3) \leq 0$
- ③ Roots 1 and 3 ; between.
- ④ $1 \leq x \leq 3$

ANSWER $[1, 3]$ **03 Interactive model**

Draw one parabola on the board and shade above the axis in one colour, below in another; every question of the lesson is a reading of that one picture.

Where is the expression positive?

$$x > 2$$



inequality $x > 2$

boundary **2 (open)**

$x = 0$ works? **no**

$x = 6$ works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Quadratic inequality	Kvadrat tengsizlik	Квадратное неравенство
Standard form	Standart ko'rinish	Стандартный вид
Interval	Oraliq	Промежуток
Union	Birlashma	Объединение
Strict inequality	Qat'iy tengsizlik	Строгое неравенство
Root of the trinomial	Uchhadning ildizi	Корень трёхчлена
Sign of the trinomial	Uchhadning ishorasi	Знак трёхчлена
Empty set	Bo'sh to'plam	Пустое множество

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

For $a > 0$, the trinomial is negative:

outside the roots

between the roots

everywhere

nowhere

$x^2 + 2x + 5 > 0$ has solution:

no x

all x

$x > 0$

two intervals

$x^2 + 2x + 5 < 0$ has solution:

all x

no x

$x < 0$

one interval

From $2x^2 \leq 5x$ you should:

divide by x

move to one side and factorise

square

take roots

Multiplying an inequality by -1 :

changes nothing

flips the sign

squares it

is not allowed

The bracket beside $+\infty$ is:

square

round

either

curly

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Standard form of $x^2 > 3x + 4$

ANSWER $x^2 - 3x - 4 > 0$

2. Roots of $x^2 - 3x - 4$

ANSWER -1 and 4

3. Solve $(x - 2)(x - 5) < 0$

ANSWER $(2, 5)$

4. Solve $(x - 2)(x - 5) > 0$

ANSWER $(-\infty, 2) \cup (5, +\infty)$

5. Solve $x^2 < 9$

ANSWER $(-3, 3)$

6. Solve $x^2 > 9$

ANSWER $(-\infty, -3) \cup (3, +\infty)$

7. Solve $x^2 \geq 0$

ANSWER All x

Medium · the standard question

7 PROBLEMS

1. Solve $x^2 - 3x - 4 > 0$

ANSWER $(-\infty, -1) \cup (4, +\infty)$

2. Solve $2x^2 \leq 5x$

ANSWER $[0, 2.5]$

3. Solve $x^2 + 2x + 5 > 0$

ANSWER All x

4. Solve $x^2 + 2x + 5 < 0$

ANSWER No solution

5. Solve $-x^2 + 4x - 3 \geq 0$

ANSWER $[1, 3]$

6. Solve $x^2 - 6x + 9 > 0$

ANSWER All $x \neq 3$

7. Solve $x^2 - 6x + 9 \leq 0$

ANSWER $x = 3$

Hard · stretch and reason

7 PROBLEMS

1. Solve $(x - 1)(x + 3) < 5$

ANSWER $(-4, 2)$

2. Solve $3x^2 - 2x - 1 \geq 0$

ANSWER $(-\infty, -\frac{1}{3}] \cup [1, +\infty)$

3. For which k is $x^2 + kx + 4 > 0$ for every x ?

ANSWER $-4 < k < 4$

4. For which m has $x^2 - 2x + m < 0$ no solution?

ANSWER $m \geq 1$

5. Solve $x^2 \leq 2x + 8$ and give the integer solutions

ANSWER $[-2, 4]; -2, \dots, 4$

6. Solve $(2x - 1)^2 > 9$

ANSWER $(-\infty, -1) \cup (2, +\infty)$

7. The domain of $\sqrt{x^2 - 4x + 3}$

ANSWER $(-\infty, 1] \cup [3, +\infty)$

07 Homework

Homework — 5 tasks

Sketch the parabola for every question; the sketch is the method, not decoration.

1. Solve $x^2 - 5x + 6 < 0$ and draw the answer on a number line.
2. Solve $x^2 + x - 12 \geq 0$.
3. Solve $3x^2 < 12x$, taking care not to divide by x .
4. Solve $x^2 - 4x + 7 > 0$ and explain the answer from the discriminant alone.

5. Find the domain of $\sqrt{9 - x^2}$.

Solving a quadratic inequality from the graph

The sketch is not an illustration of the method — it is the method.

LESSON 11–12

2 × 40 MIN

ALGEBRA 9, §7

IGX 10.2

LESSON CLOCK

13'

20'

23'

20'

4'

Reading the sketch	13 min
A curve against a line	20 min
Always true, never true	23 min
Checking an answer	20 min
Homework	4 min

LEARNING OBJECTIVES

- Read the solution of an inequality directly from a sketched parabola.
- Solve an inequality where one side is a line rather than zero.
- Find the values of a parameter for which an inequality always holds.
- Use the graph to check an algebraic answer.

TEXTBOOK

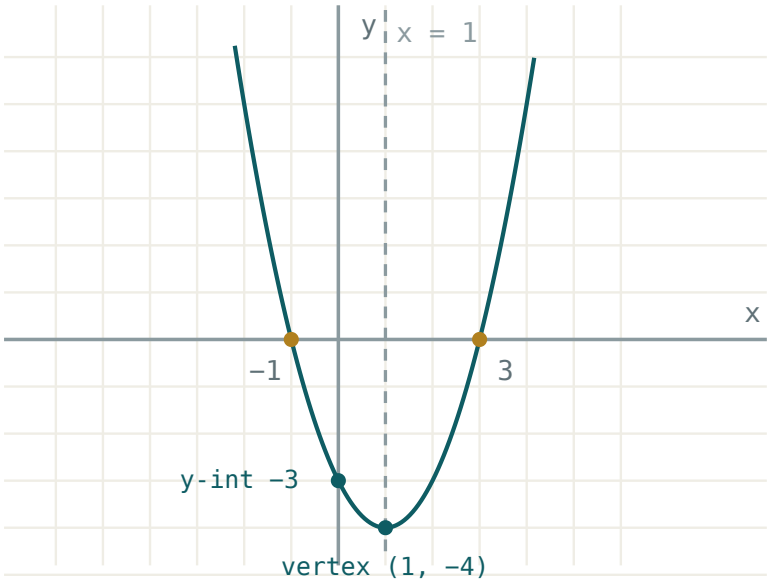
National	pp. 28–31
Cambridge	Core & Extended, pp. 216–225
Workbook	Exercise 10.2

01

Explanation

Reading the sketch

$f(x) > 0$ asks where the curve is **above** the axis; $f(x) < 0$ asks where it is **below**.
Nothing else is needed.



Above the axis on the outside, below it between the roots.

THREE MARKS, THEN READ

1. Mark the roots on the x -axis.
2. Sketch the parabola through them, opening the way a says.
3. Trace the part above (or below) the axis and read off the x values.

A sketch takes fifteen seconds and removes every sign error at once.

A curve against a line

$x^2 < 2x + 3$ can be read two ways, and both are useful.

READING	QUESTION	WORKING
one curve	where is $x^2 - 2x - 3$ below zero?	roots $-1, 3$; between
two graphs	where is the parabola below the line?	same roots; between

The second reading is the one to use when the question is already drawn — a parabola and a line on the same axes, and “for which x is the curve below the line?”.

The meeting points are the roots of the difference, and the answer is the interval between or outside them.

THE MEETING POINTS ARE ALWAYS THE ROOTS OF THE DIFFERENCE

Curve $y = f(x)$ meets line $y = g(x)$ where $f(x) - g(x) = 0$. So the graphical question and the algebraic one have exactly the same answer — which is why the sketch is a legitimate solution and not just a check.

Always true, never true

Some questions ask for no x at all, but for the values of a letter.

CONDITION	MEANS	REQUIRES
$ax^2 + bx + c > 0$	for every x	$a > 0$ and $D < 0$
$ax^2 + bx + c < 0$	for every x	$a < 0$ and $D < 0$
$ax^2 + bx + c \geq 0$	for every x	$a > 0$ and $D \leq 0$

Example. For which k is $x^2 + kx + 9 > 0$ for every x ? Here $a = 1 > 0$ already, so the condition is $D < 0$: $k^2 - 36 < 0$, that is $-6 < k < 6$.

TWO CONDITIONS, NOT ONE

$D < 0$ alone says the curve misses the axis — but it could be entirely **below** it. The sign of a is what says which. Write both, always.

Checking an answer

Every inequality answer can be tested in ten seconds: pick one number from the claimed set and one from outside it, and substitute.

CLAIM	TEST INSIDE	TEST OUTSIDE	VERDICT
$x^2 - 3x - 4 > 0$ on $(4, \infty)$	$x = 5: 6 > 0 \checkmark$	$x = 0: -4 > 0 \text{ ✗}$	consistent
$x^2 < 9$ on $(-3, 3)$	$x = 0: 0 < 9 \checkmark$	$x = 4: 16 < 9 \text{ ✗}$	consistent

TEST THE ENDPOINTS TOO

For a weak inequality, substitute the endpoint itself: $x = 4$ in $x^2 - 3x - 4 \geq 0$ gives $0 \geq 0 \checkmark$, so 4 belongs and the bracket is square. That single substitution settles every open-or-closed question.

02 Worked examples

EXAMPLE 1 Solve $x^2 < 2x + 3$ graphically.

1 Sketch $y = x^2$ and $y = 2x + 3$.

2 They meet where $x^2 - 2x - 3 = 0$, so at $x = -1$ and $x = 3$.

3 The parabola is below the line between them.

4 $-1 < x < 3$

ANSWER

$(-1, 3)$

EXAMPLE 2 For which k is $x^2 + kx + 9 > 0$ for every x ?

① $a = 1 > 0$ ✓

First condition already met.

② $D = k^2 - 36 < 0$

③ $k^2 < 36$

④ $-6 < k < 6$

ANSWER

$$-6 < k < 6$$

EXAMPLE 3 For which m is $-x^2 + mx - 4 < 0$ for every x ?

① $a = -1 < 0$ ✓

② $D = m^2 - 16 < 0$

③ $-4 < m < 4$

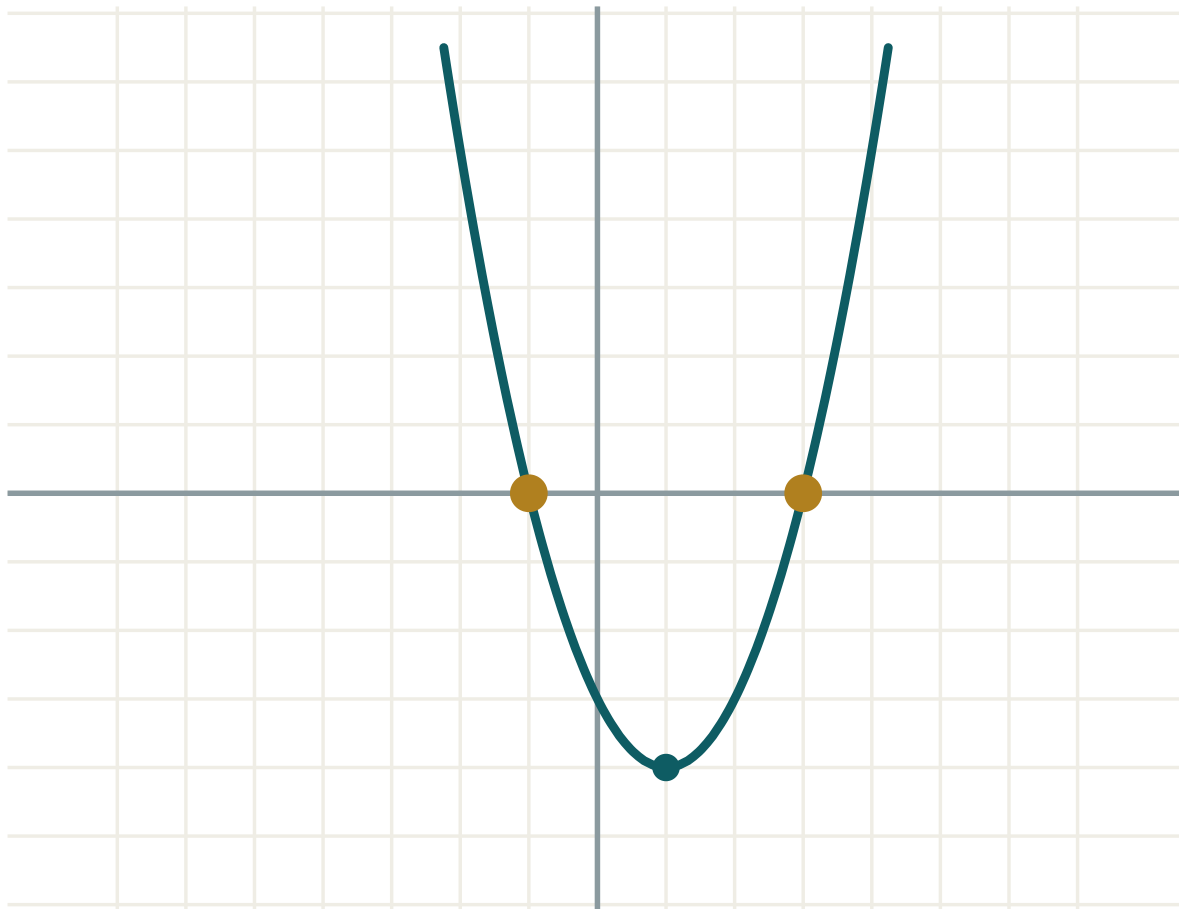
ANSWER

$$-4 < m < 4$$

03 Interactive model

Draw a parabola and a movable line on the board; slide the line and ask each time which x satisfy “curve below line”.

The curve, the axis and the shaded set

a **1**b **-2**c **-3** $D = b^2 - 4ac$ **16**roots **two** x_1 **-1** x_2 **3** $x_1 + x_2$ **2** $x_1 \cdot x_2$ **-3**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Graphical method	Grafik usul	Графический способ
Above the axis	O'q ustida	Выше оси
Below the axis	O'q ostida	Ниже оси
Point of intersection	Kesishish nuqtasi	Точка пересечения
Parameter	Parametr	Параметр
Always true	Doimo o'rinli	Всегда верно
Sketch	Eskiz	Эскиз
Solution set	Yechimlar to'plami	Множество решений

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$f(x) > 0$ asks where the curve is:

above the axis

below the axis

steepest

at its vertex

A curve meets a line where:

$f(x) = 0$

$f(x) - g(x) = 0$

$g(x) = 0$

never

$ax^2 + bx + c > 0$ for all x needs:

$D < 0$

$a > 0$

both

neither

$x^2 + kx + 9 > 0$ always, when:

$k > 6$

$-6 < k < 6$

$k < -6$

any k

To test an interval answer:

re-solve it

substitute one point inside and one outside

draw it again

nothing

A weak inequality includes an endpoint when:

always

substituting it gives a true statement

never

the root is an integer

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Where is $y = x^2 - 4$ above Ox ?

ANSWER $(-\infty, -2) \cup (2, +\infty)$

2. Where is $y = x^2 - 4$ below Ox ?

ANSWER $(-2, 2)$

3. Where do $y = x^2$ and $y = 4$ meet?

ANSWER $x = \pm 2$

4. Where do $y = x^2$ and $y = 2x + 3$ meet?

ANSWER $x = -1, 3$

5. Solve $x^2 < 4$

ANSWER $(-2, 2)$

6. Solve $x^2 \geq 4$

ANSWER $(-\infty, -2] \cup [2, +\infty)$

7. Does $x = 4$ satisfy $x^2 - 3x - 4 \geq 0$?

ANSWER Yes — $0 \geq 0$

Medium · the standard question

7 PROBLEMS

1. Solve $x^2 < 2x + 3$

ANSWER $(-1, 3)$

2. Solve $x^2 \geq 4x - 3$

ANSWER $(-\infty, 1] \cup [3, +\infty)$

3. For which k is $x^2 + kx + 9 > 0$ always?

ANSWER $-6 < k < 6$

4. For which m is $-x^2 + mx - 4 < 0$ always?

ANSWER $-4 < m < 4$

5. Where is $y = -x^2 + 4$ above Ox ?

ANSWER $(-2, 2)$

6. Where is the parabola $y = x^2$ below $y = x + 6$?

ANSWER $(-2, 3)$

7. Solve $x^2 + 4 > 0$

ANSWER All x

Hard · stretch and reason

7 PROBLEMS

1. For which a is $ax^2 + 4x + 1 > 0$ for every x ?

ANSWER $a > 4$

2. For which k does $y = x^2 + k$ lie entirely above $y = 2x$?

ANSWER $k > 1$

3. Solve $(x - 1)^2 \geq 4$

ANSWER $(-\infty, -1] \cup [3, +\infty)$

4. For which m has $x^2 - mx + m = 0$ two distinct roots?

ANSWER $m < 0$ or $m > 4$

5. Solve $x^2 - 5|x| + 6 < 0$

ANSWER $(-3, -2) \cup (2, 3)$

6. The parabola $y = x^2 - 6x + c$ lies entirely above Ox : find c

ANSWER $c > 9$

7. Solve $\frac{1}{x} < x$ for $x > 0$

ANSWER $x > 1$

07 Homework

Homework — 5 tasks

Every answer must be tested with one number inside and one outside the claimed set.

1. Solve $x^2 \leq 3x + 10$ graphically, drawing both the parabola and the line.
2. Solve $x^2 - 2x - 8 > 0$ and test your answer at two points.
3. For which k is $x^2 + kx + 16 > 0$ for every x ?
4. For which c does $y = x^2 - 4x + c$ lie entirely above the x -axis?

5. Explain in two sentences why $D < 0$ alone is not enough for $f(x) > 0$ to hold everywhere.

The method of intervals

One number line, one sign at the right-hand end, and the rest alternate.

LESSON 13–14

2 × 40 MIN

ALGEBRA 9, §8

EXTENSION BEYOND IGX

LESSON CLOCK

13'

23'

20'

20'

4'

Why it works	13 min
The four steps	23 min
Repeated factors	20 min
Rational inequalities	20 min
Homework	4 min

LEARNING OBJECTIVES

- Factorise a polynomial or rational expression completely.
- Mark the zeros and the breaks on a number line.
- Determine the sign on each interval and read off the answer.
- Handle a repeated factor, where the sign does not change.

TEXTBOOK

National	pp. 32–36
Cambridge	Core & Extended, pp. 216–225
Workbook	Exercise 10.2

01

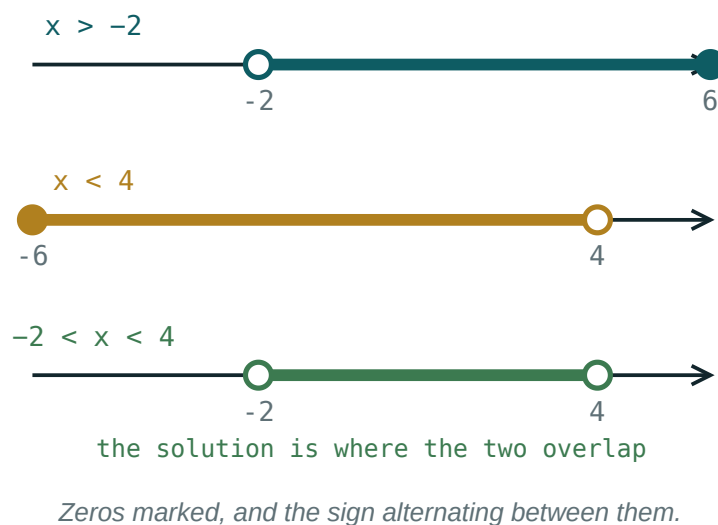
Explanation

Why it works

A product of factors can only change sign where one of the factors is zero. Between two consecutive zeros, every factor keeps its sign, so the product does too.

the sign is constant on each interval between consecutive zeros

So the whole number line splits into a few intervals, and one test point in each settles it. Better still: crossing a **single** zero flips exactly one factor, so the sign **alternates** — and one test point settles them all.



THIS REPLACES THE PARABOLA SKETCH, AND GOES FURTHER

For a quadratic, the method of intervals and the sketch give the same answer. For a cubic, a quartic or a fraction, the sketch is hard and the number line is not — which is why this method is the one that survives into Grade 10 and 11.

The four steps

EVERY QUESTION, THE SAME FOUR STEPS

1. **Zero on one side**, and factorise completely.
2. **Mark** every zero on a number line — filled for $\leq, \geq$, open for $<, >$.
3. **Sign at the far right**: substitute a large x , or read the leading coefficient.
4. **Alternate** leftwards through the simple zeros, and read off the intervals wanted.

Example. $(x + 2)(x - 1)(x - 4) > 0$.

INTERVAL	$x < -2$	$-2 \dots 1$	$1 \dots 4$	$x > 4$
sign	–	+	–	+

The far right is + (three positive factors), and the signs alternate leftwards. We want > 0 , so the answer is $(-2, 1) \cup (4, +\infty)$.

Repeated factors

A factor to an **even** power does not change sign as x passes its zero — $(x - 3)^2$ is positive on both sides. An **odd** power behaves like a single factor.

FACTOR	AT ITS ZERO	SIGN
$(x - 3)$	crosses	changes
$(x - 3)^2$	touches	does not change
$(x - 3)^3$	crosses	changes
$(x - 3)^4$	touches	does not change

DO NOT CANCEL A SQUARE — MARK IT AND SKIP THE FLIP

In $(x - 1)(x - 3)^2 > 0$ the point $x = 3$ is still excluded (the product is zero there), but the sign is $+$ on both sides of it. The answer is $(1, 3) \cup (3, +\infty)$ — one interval with a hole in it, not two separate regions.

Rational inequalities

A fraction changes sign at the zeros of its **numerator** and at the zeros of its **denominator**. Mark both — but the denominator's zeros are always excluded, whatever the inequality sign.

$\frac{P(x)}{Q(x)} > 0$ has the same solution as $P(x) \cdot Q(x) > 0$, with $Q(x) \neq 0$

POINT	CIRCLE	BECAUSE
zero of the numerator, strict	open	the fraction is 0 , not > 0
zero of the numerator, weak	filled	$0 \geq 0$
zero of the denominator	always open	the fraction is undefined

NEVER MULTIPLY UP BY THE DENOMINATOR

From $\frac{x - 1}{x + 2} > 0$ you may **not** multiply by $x + 2$ — its sign is unknown. Bring everything to one fraction, factorise, and use the number line.

02 Worked examples

EXAMPLE 1 Solve $(x + 2)(x - 1)(x - 4) > 0$.

- 1 Zeros $-2, 1, 4$, all simple.
- 2 Far right: $+$.
Three positive factors.
- 3 Signs, right to left: $+, -, +, -$.
- 4 $(-2, 1) \cup (4, +\infty)$

ANSWER

$(-2, 1) \cup (4, +\infty)$

EXAMPLE 2 Solve $(x - 1)(x - 3)^2 > 0$.

- 1 Zeros 1 (simple) and 3 (double).
- 2 Far right $+$; no flip at 3 ; flip at 1 .
- 3 Signs: $-$ then $+$ then $+$.
- 4 $(1, 3) \cup (3, +\infty)$
 $x = 3$ makes the product zero.

ANSWER

$(1, 3) \cup (3, +\infty)$

EXAMPLE 3 Solve $\frac{x-1}{x+2} \geq 0$.

① Zeros: numerator at 1, denominator at -2.

② $x = 1$ filled; $x = -2$ always open.

③ Far right +; alternate leftwards.

④ $(-\infty, -2) \cup [1, +\infty)$

ANSWER

$(-\infty, -2) \cup [1, +\infty)$

EXAMPLE 4 Solve $\frac{1}{x} < 3$.

① $\frac{1}{x} - 3 < 0$

Never multiply up.

② $\frac{1-3x}{x} < 0$

③ Zeros $\frac{1}{3}$ (open) and 0 (open).

④ $(-\infty, 0) \cup (\frac{1}{3}, +\infty)$

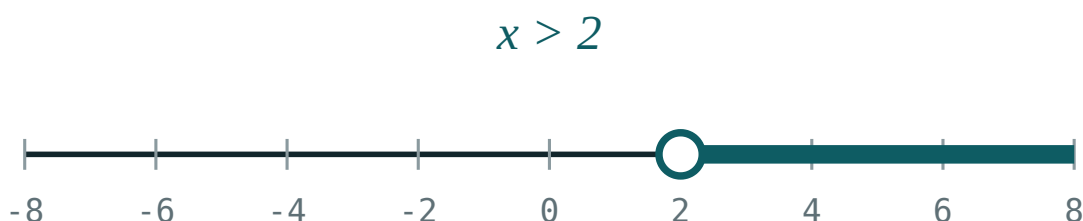
ANSWER

$(-\infty, 0) \cup (\frac{1}{3}, +\infty)$

03 Interactive model

Draw one long number line on the board and let a pupil place the circles before anyone writes a sign.

Zeros, signs and the answer



inequality $x > 2$

boundary **2 (open)**

$x = 0$ works? **no**

$x = 6$ works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Method of intervals	Oraliqlar usuli	Метод интервалов
Zero of a factor	Ko'paytuvchi noli	Ноль множителя
Break point	Uzilish nuqtasi	Точка разрыва
Repeated factor	Takrorlanuvchi ko'paytuvchi	Кратный множитель
Multiplicity	Karralilik	Кратность
Sign chart	Ishoralar chizmasi	Схема знаков
Rational inequality	Ratsional tengsizlik	Рациональное неравенство
Excluded value	Chetlangan qiymat	Исключённое значение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A product can change sign only:

at a zero of a factor

anywhere

at the origin

never

Crossing a simple zero, the sign:

stays

flips

doubles

vanishes

Crossing a double zero, the sign:

stays

flips

is undefined

is zero

A zero of the denominator is drawn:

From $\frac{x-1}{x+2} > 0$ you may:

The sign at the far right of $(x+2)(x-1)(x-4)$ is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Zeros of $(x + 2)(x - 1)(x - 4)$

ANSWER $-2, 1, 4$

2. Sign of that product at $x = 10$

ANSWER $+$

3. Solve $(x - 1)(x - 5) < 0$

ANSWER $(1, 5)$

4. Solve $(x - 1)(x - 5) > 0$

ANSWER $(-\infty, 1) \cup (5, +\infty)$

5. Solve $x(x - 3) \leq 0$

ANSWER $[0, 3]$

6. Excluded value of $\frac{1}{x - 4}$

ANSWER $x = 4$

7. Does $(x - 3)^2$ change sign at 3?

ANSWER No

Medium · the standard question

7 PROBLEMS

1. Solve $(x + 2)(x - 1)(x - 4) > 0$

ANSWER $(-2, 1) \cup (4, +\infty)$

2. Solve $(x + 2)(x - 1)(x - 4) < 0$

ANSWER $(-\infty, -2) \cup (1, 4)$

3. Solve $(x - 1)(x - 3)^2 > 0$

ANSWER $(1, 3) \cup (3, +\infty)$

4. Solve $\frac{x - 1}{x + 2} \geq 0$

ANSWER $(-\infty, -2) \cup [1, +\infty)$

5. Solve $\frac{x + 3}{x - 1} < 0$

ANSWER $(-3, 1)$

6. Solve $x^3 - 4x > 0$

ANSWER $(-2, 0) \cup (2, +\infty)$

7. Solve $\frac{1}{x} < 3$

ANSWER $(-\infty, 0) \cup (\frac{1}{3}, +\infty)$
Hard · stretch and reason

7 PROBLEMS

1. Solve $\frac{(x-1)(x+3)}{x-2} \leq 0$

ANSWER $(-\infty, -3] \cup [1, 2)$

2. Solve $(x-1)^2(x+2)(x-5) < 0$

ANSWER $(-2, 1) \cup (1, 5)$

3. Solve $\frac{x^2-4}{x^2-9} > 0$

ANSWER $(-\infty, -3) \cup (-2, 2) \cup (3, +\infty)$

4. Solve $\frac{2}{x-1} \geq \frac{3}{x}$

ANSWER $(0, 1) \cup [3, +\infty)$ — check the ends

5. Solve $x^4 - 5x^2 + 4 \leq 0$

ANSWER $[-2, -1] \cup [1, 2]$

6. The domain of $\sqrt{\frac{x-1}{x+2}}$

ANSWER $(-\infty, -2) \cup [1, +\infty)$

7. Solve $(x-2)^3(x+1) > 0$

ANSWER $(-\infty, -1) \cup (2, +\infty)$

07 Homework

Homework — 5 tasks

Draw the number line for every question, with the circles filled or open correctly.

1. Solve $(x+1)(x-2)(x-5) < 0$.

2. Solve $(x+4)(x-2)^2 \geq 0$, explaining what happens at $x = 2$.

3. Solve $\frac{x-3}{x+1} > 0$.

4. Solve $\frac{2}{x} \leq 1$ without multiplying up.

5. Find the domain of $\sqrt{x^2 - x - 6}$.

Control work 1, and work on the mistakes

The quadratic function and its inequalities in one paper, then a named diagnosis of each slip.

LESSON 15–16

2 × 40 MIN

ALGEBRA 9, NAZORAT ISHI 1

IGX 10 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Sketch a quadratic and read its features under time.
- Solve a quadratic inequality and a simple rational one.
- Classify each lost mark and rewrite the solution in full.
- Draw the chapter so far as one map.

TEXTBOOK

National	pp. 56–62
Cambridge	Core & Extended, pp. 226–228
Workbook	Control paper A1

01

Explanation

The paper — 30 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Write $y = x^2 - 8x + 11$ in vertex form; state the vertex and the least value	5	L5–6
2	Sketch $y = x^2 - 2x - 8$: vertex, axis, both roots and the intercept	6	L7–8
3	Solve $x^2 - 2x - 8 \geq 0$ and write the answer as intervals	5	L9–10
4	Solve $x^2 + 3x + 7 > 0$, justifying the answer from D	4	L11–12
5	Solve $(x + 1)(x - 2)(x - 4) < 0$ by the method of intervals	5	L13–14
6	Solve $\frac{x - 2}{x + 3} \leq 0$	5	L13–14

WHERE THE MARKS ACTUALLY GO

Q2 carries one mark for the axis being halfway between the roots; Q3 one for the square brackets; Q4 one for naming **both** conditions, not just $D < 0$; Q6 one for the open circle at $x = -3$. Four of the thirty

marks are for a habit rather than a calculation.

Naming the slip

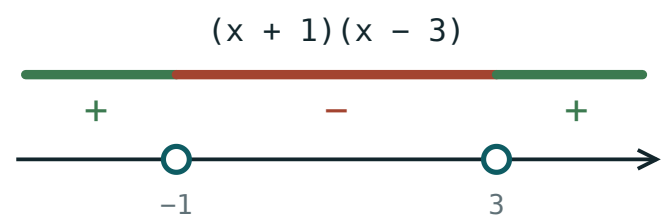
SLIP	WHAT IT LOOKS LIKE	THE FIX
bracket sign reversed	$(x - 3)^2 \Rightarrow \text{vertex at } -3$	read as $(x - (+3))^2$
outside for between	the wrong side of the roots	sketch, then read
open for closed	$(1, 4)$ for $x^2 - 5x + 4 \leq 0$	substitute the endpoint
$D < 0$ alone	“no roots, so always positive”	the sign of a decides which side
denominator filled	$[-3, 2]$	it is never allowed
multiplied up	$x - 2 \leq 0$	one fraction, then intervals
divided by x	$2x^2 \leq 5x \Rightarrow 2x \leq 5$	factorise instead

Write the slip's name in the margin of every question that lost a mark, then rewrite the whole solution — not the line that went wrong.

The chapter so far, as one map

Five boxes, and the links written as sentences:

- $a \rightarrow$ **the direction** — “up when positive, down when negative”
- **completing the square** $\rightarrow$ **the vertex** — “ $a(x - m)^2 + n$ puts it at (m, n) ”
- $D \rightarrow$ **how many roots** — “two, one, or none”
- **the roots** $\rightarrow$ **the sign** — “ a 's sign outside, the opposite between”
- **factors** $\rightarrow$ **the number line** — “mark, alternate, read”



negative between the roots, positive outside

The one picture that answers Q3, Q4 and Q5 alike.

LOOKING FORWARD

Lessons 17–23 leave the graph and study the **function** itself — its domain, where it rises and falls, and whether it is even or odd. Everything drawn this fortnight becomes the evidence for those statements.

02 Worked examples

EXAMPLE 1 Model answer, Q2: sketch $y = x^2 - 2x - 8$.

- 1 $a = 1 > 0$; intercept $(0, -8)$.
- 2 $(x - 4)(x + 2) = 0$, so roots -2 and 4 .
- 3 Axis $x = 1$.
Halfway between the roots.
- 4 Vertex $(1, -9)$.

ANSWER

Vertex $(1, -9)$, roots -2 and 4

EXAMPLE 2 Model answer, Q4: solve $x^2 + 3x + 7 > 0$.

① $D = 9 - 28 = -19 < 0$

No roots.

② $a = 1 > 0$

So the curve is entirely above Ox .

③ Both conditions are needed.

④ The inequality holds for every real x .

ANSWER

All real x

EXAMPLE 3 Model answer, Q6: solve $\frac{x-2}{x+3} \leq 0$.

① Zeros: 2 from the top, -3 from the bottom.

② $x = 2$ filled; $x = -3$ open, always.

③ Far right +; alternate leftwards.

④ $(-3, 2]$

ANSWER

$(-3, 2]$

03 Interactive model

Work Q6 twice on the board — once with the -3 filled, once open — and let the class say which is a solution of the original fraction.

The chapter in ten questions

$y = ax^2 + bx + c$ is quadratic when:

$$a \neq 0$$

$$b \neq 0$$

$$c \neq 0$$

always

The vertex of $y = (x - 5)^2 + 2$:

$$(5, 2)$$

$$(-5, 2)$$

$$(5, -2)$$

$$(2, 5)$$

The axis of $y = ax^2 + bx + c$:

$$x = \frac{b}{2a}$$

$$x = -\frac{b}{2a}$$

$$x = c$$

$$x = 0$$

$D = 0$ means the parabola:

cuts twice

touches

misses

is a line

For $a > 0$, the trinomial is negative:

outside the roots

between them

everywhere

nowhere

$x^2 + 4 > 0$ holds for:

no x

all x

$x > 0$

$x > 2$

A double zero makes the sign:

flip

stay

vanish

double

A zero of the denominator is:

filled

open

ignored

doubled

From $\frac{1}{x} < 3$ you should:

multiply by x

bring to one fraction

square

take reciprocals

The bracket beside $+\infty$:

square

round

either

none

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Vertex	Uchi	Вершина
Discriminant	Diskriminant	Дискриминант
Method of intervals	Oraliqlar usuli	Метод интервалов
Sign chart	Ishoralar chizmasi	Схема знаков
Diagnosis	Tashxis	Диагностика
Concept map	Tushunchalar xaritasi	Карта понятий
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q3's answer uses which brackets?

round

square at the roots

curly

none

Q4 needs, beyond $D < 0$:

nothing

the sign of a

the roots

a sketch

Q6's left endpoint is:

filled

open

either

excluded from the answer entirely

Lessons 17–23 turn to:

trigonometry

the properties of functions

progressions

probability

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Vertex form of $y = x^2 - 8x + 11$

ANSWER $(x - 4)^2 - 5$

2. Vertex of the same

ANSWER $(4, -5)$

3. Roots of $x^2 - 2x - 8$

ANSWER -2 and 4

4. Axis of $y = x^2 - 2x - 8$

ANSWER $x = 1$

5. Vertex of $y = x^2 - 2x - 8$

ANSWER $(1, -9)$

6. D of $x^2 + 3x + 7$

ANSWER -19

7. Zeros of $(x + 1)(x - 2)(x - 4)$

ANSWER $-1, 2, 4$

Medium · the standard question

7 PROBLEMS

1. Solve $x^2 - 2x - 8 \geq 0$

ANSWER $(-\infty, -2] \cup [4, +\infty)$

2. Solve $x^2 - 2x - 8 < 0$

ANSWER $(-2, 4)$

3. Solve $x^2 + 3x + 7 > 0$

ANSWER All x

4. Solve $x^2 + 3x + 7 < 0$

ANSWER No solution

5. Solve $(x + 1)(x - 2)(x - 4) < 0$

ANSWER $(-\infty, -1) \cup (2, 4)$

6. Solve $\frac{x - 2}{x + 3} \leq 0$

ANSWER $(-3, 2]$

7. Least value of $y = x^2 - 8x + 11$

ANSWER -5

Hard · stretch and reason

7 PROBLEMS

1. Solve $(x + 1)(x - 2)^2(x - 4) \leq 0$

ANSWER $[-1, 4]$

2. Solve $\frac{x^2 - 4}{x - 1} \geq 0$

ANSWER $[-2, 1) \cup [2, +\infty)$

3. For which k is $x^2 + kx + 7 > 0$ always?

ANSWER $-2\sqrt{7} < k < 2\sqrt{7}$

4. Solve $x^4 - 10x^2 + 9 < 0$

ANSWER $(-3, -1) \cup (1, 3)$

5. The domain of $\sqrt{x^2 - 2x - 8}$

ANSWER $(-\infty, -2] \cup [4, +\infty)$

6. Solve $\frac{3}{x + 1} > \frac{2}{x}$

ANSWER $(-1, 0) \cup (2, +\infty)$

7. A parabola with least value -9 and roots $-2, 4$

ANSWER $y = x^2 - 2x - 8$

07 Homework

Homework — 4 tasks

Bring the concept map to Lesson 17; the properties of functions are read off it.

1. Rewrite in full every control-work question that lost a mark, naming the slip in the margin.
2. Complete the five-box concept map with each link written as a sentence.
3. Solve $\frac{x + 4}{x - 1} \geq 0$ and draw the answer on a number line.

4. Write your target for the rest of Chapter I in one checkable sentence, and date it.

The domain of a function

Three things forbid a value: a zero denominator, a negative under a square root, and both at once.

LESSON 17–18

2 × 40 MIN

ALGEBRA 9, §9

IGX 10.1

LESSON CLOCK

11'

20'

22'

20'

7'

What a domain is	11 min
The two forbidden things	20 min
Roots and quadratics	22 min
Both at once	20 min
Homework	7 min

LEARNING OBJECTIVES

- State what the domain of a function means.
- Find the domain of a rational, a root and a mixed expression.
- Write a domain as an interval or a union of intervals.
- Recognise when a domain question is really a quadratic inequality.

TEXTBOOK

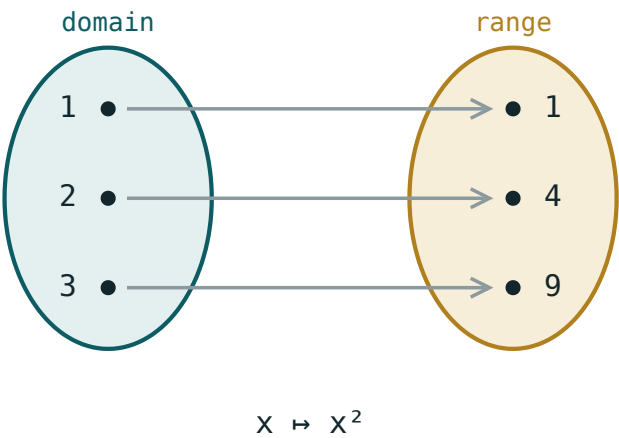
National	pp. 37–40
Cambridge	Core & Extended, pp. 198–215
Workbook	Exercise 10.1

01

Explanation

What a domain is

The **domain** $D(f)$ is the set of x for which the formula makes sense. Unless a problem says otherwise, it is the largest such set — the **natural** domain.



Every x of the domain sends one y to the range; a forbidden x has no arrow at all.

FUNCTION	DOMAIN	WHY
$y = 3x - 7$	all x	nothing forbids anything
$y = x^2 + 1$	all x	polynomials are always defined
$y = \frac{1}{x}$	$x \neq 0$	no division by zero
$y = \sqrt{x}$	$x \geq 0$	no root of a negative

POLYNOMIALS ARE FREE

Any expression built only from $+$, $-$, $\times$ and whole-number powers is defined everywhere. Only a **division** or a **square root** can forbid a value — so look for those two and nothing else.

The two forbidden things

denominator $\neq 0$ · radicand ≥ 0

FUNCTION	CONDITION	DOMAIN
$\frac{1}{x-3}$	$x-3 \neq 0$	$x \neq 3$
$\frac{x}{x^2-4}$	$x^2-4 \neq 0$	$x \neq \pm 2$
$\sqrt{x-5}$	$x-5 \geq 0$	$[5, +\infty)$
$\sqrt{7-x}$	$7-x \geq 0$	$(-\infty, 7]$

A ROOT IN THE DENOMINATOR OBEYS BOTH RULES AT ONCE

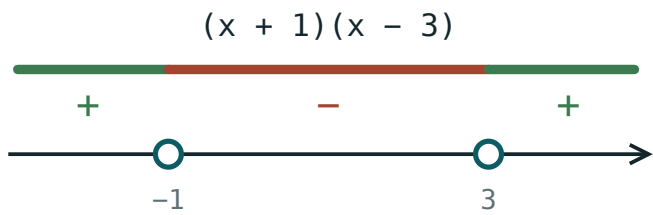
For $\frac{1}{\sqrt{x-5}}$ the radicand must be non-negative **and** the denominator non-zero, so $x-5 > 0$ — strictly. The domain is $(5, +\infty)$, not $[5, +\infty)$.

Roots and quadratics

When the radicand is a quadratic, the domain question **is** a quadratic inequality — exactly the work of Lessons 9–14.

Example. $y = \sqrt{x^2 - 5x + 6}$. Need $x^2 - 5x + 6 \geq 0$, that is $(x - 2)(x - 3) \geq 0$: outside the roots, inclusive.

$D(f) = (-\infty, 2] \cup [3, +\infty)$



negative between the roots, positive outside

The same sign chart, now answering a domain question.

FUNCTION	INEQUALITY	DOMAIN
$\sqrt{9 - x^2}$	$9 - x^2 \geq 0$	$[-3, 3]$
$\sqrt{x^2 + 1}$	$x^2 + 1 \geq 0$	all x
$\sqrt{-x^2 - 1}$	$-x^2 - 1 \geq 0$	$\emptyset$

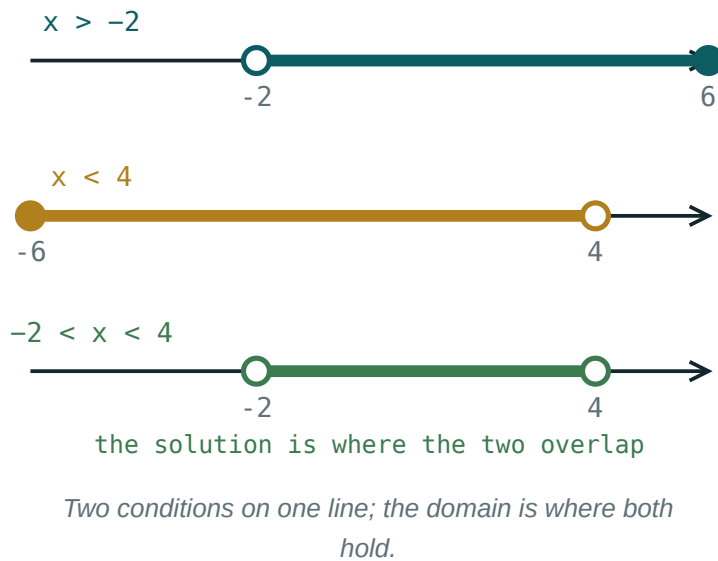
The last two are the degenerate cases again: $x^2 + 1$ is never negative, so nothing is forbidden; $-x^2 - 1$ is never positive, so **everything** is.

Both at once

When several conditions appear, find each and take the **intersection** — the values that satisfy all of them.

Example. $y = \frac{\sqrt{x - 1}}{x - 4}$.

SOURCE	CONDITION	SET
the root	$x - 1 \geq 0$	$[1, +\infty)$
the denominator	$x - 4 \neq 0$	$x \neq 4$
together	both	$[1, 4) \cup (4, +\infty)$

**DRAW BOTH ON THE SAME LINE**

Mark the first condition above the line and the second below it. The domain is the part where both marks agree — which is far quicker, and far safer, than trying to intersect the intervals in your head.

02 Worked examples

EXAMPLE 1 Find the domain of $y = \frac{x+1}{x^2-9}$.

- ① $x^2 - 9 \neq 0$
Only the denominator can forbid.

② $(x-3)(x+3) \neq 0$

③ $x \neq 3$ and $x \neq -3$

④ $(-\infty, -3) \cup (-3, 3) \cup (3, +\infty)$

ANSWER

All x except ± 3

EXAMPLE 2 Find the domain of $y = \sqrt{x^2 - 5x + 6}$.

① $x^2 - 5x + 6 \geq 0$

② $(x - 2)(x - 3) \geq 0$

③ $a > 0$, want ≥ 0 — outside, inclusive.

④ $(-\infty, 2] \cup [3, +\infty)$

ANSWER

$(-\infty, 2] \cup [3, +\infty)$

EXAMPLE 3 Find the domain of $y = \frac{\sqrt{x-1}}{x-4}$.

① $x - 1 \geq 0 \Rightarrow x \geq 1$

② $x - 4 \neq 0 \Rightarrow x \neq 4$

③ Intersect the two.

④ $[1, 4) \cup (4, +\infty)$

ANSWER

$[1, 4) \cup (4, +\infty)$

EXAMPLE 4 Find the domain of $y = \frac{1}{\sqrt{4 - x^2}}$.

① A root downstairs: $4 - x^2 > 0$, strictly.

② $x^2 < 4$

③ $-2 < x < 2$

ANSWER

$(-2, 2)$

03 Interactive model

Write five formulas on the board and ask only “what could possibly go wrong here?” before anyone computes.

Where is the expression defined?

$$x > 2$$



inequality $x > 2$

boundary 2 (open)

$x = 0$ works? no

$x = 6$ works? yes

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Domain	Aniqlanish sohasi	Область определения
Range	Qiymatlar sohasi	Область значений
Denominator	Maxraj	Знаменатель
Radicand	Ildiz ostidagi ifoda	Подкоренное выражение
Admissible value	Mumkin bo'lgan qiymat	Допустимое значение
Excluded value	Chetlangan qiymat	Исключённое значение
Union of intervals	Oraliqlar birlashmasi	Объединение промежутков
Natural domain	Tabiiy soha	Естественная область

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The domain of a polynomial is:

$x \geq 0$

all x

$x \neq 0$

an interval

Only two things forbid a value:

addition and subtraction

a zero denominator and a negative radicand

squares and cubes

brackets

The domain of $\sqrt{x - 5}$ is:

$x > 5$

$x \geq 5$

$x \leq 5$

all x

The domain of $\sqrt{x-5}$ is:

The domain of $\sqrt{x^2+1}$ is:

Several conditions are combined by:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Domain of $y = 3x - 7$

ANSWER All x

2. Domain of $y = \frac{1}{x}$

ANSWER $x \neq 0$

3. Domain of $y = \frac{1}{x - 3}$

ANSWER $x \neq 3$

4. Domain of $y = \sqrt{x}$

ANSWER $[0, +\infty)$

5. Domain of $y = \sqrt{x - 5}$

ANSWER $[5, +\infty)$

6. Domain of $y = \sqrt{7 - x}$

ANSWER $(-\infty, 7]$

7. Domain of $y = x^2 + 1$

ANSWER All x

Medium · the standard question

7 PROBLEMS

1. Domain of $y = \frac{x+1}{x^2-9}$

ANSWER All x except ± 3

2. Domain of $y = \sqrt{x^2 - 5x + 6}$

ANSWER $(-\infty, 2] \cup [3, +\infty)$

3. Domain of $y = \sqrt{9 - x^2}$

ANSWER $[-3, 3]$

4. Domain of $y = \frac{1}{\sqrt{x-5}}$

ANSWER $(5, +\infty)$

5. Domain of $y = \frac{\sqrt{x-1}}{x-4}$

ANSWER $[1, 4) \cup (4, +\infty)$

6. Domain of $y = \frac{1}{\sqrt{4-x^2}}$

ANSWER $(-2, 2)$

7. Domain of $y = \sqrt{-x^2 - 1}$

ANSWER $\emptyset$

Hard · stretch and reason

7 PROBLEMS

1. Domain of $y = \sqrt{x} + \sqrt{6-x}$

ANSWER $[0, 6]$

2. Domain of $y = \frac{\sqrt{x+2}}{x^2 - x - 6}$

ANSWER $[-2, 3) \cup (3, +\infty)$

3. Domain of $y = \sqrt{\frac{x-1}{x+2}}$

ANSWER $(-\infty, -2) \cup [1, +\infty)$

4. Domain of $y = \frac{1}{x^2 - 5x + 6}$

ANSWER All x except 2 and 3

5. Domain of $y = \sqrt{x^2 - 4} + \frac{1}{x}$

ANSWER $(-\infty, -2] \cup [2, +\infty)$

6. For which k is the domain of $\sqrt{x^2 + kx + 4}$ all of $\mathbb{R}$?

ANSWER $-4 \leq k \leq 4$

7. Domain of $y = \frac{\sqrt{4-x}}{\sqrt{x-1}}$

ANSWER $(1, 4]$

07 Homework

Homework — 5 tasks

Write the condition before the answer; the mark is for the condition.

1. Find the domain of $y = \frac{2x}{x^2 - 16}$.

2. Find the domain of $y = \sqrt{x^2 - x - 12}$.
3. Find the domain of $y = \frac{\sqrt{x+3}}{x-2}$.
4. Find the domain of $y = \frac{1}{\sqrt{16-x^2}}$.
5. Explain in two sentences why the domain of $\frac{1}{\sqrt{x}}$ is $(0, +\infty)$ and not $[0, +\infty)$.

Increasing and decreasing functions

Read the graph left to right: uphill is increasing, downhill is decreasing.

LESSON 19

1 × 40 MIN

ALGEBRA 9, §10

IGX 10.1

LESSON CLOCK

8'

12'

12'

8'

The definition	8 min
Reading a graph	12 min
Quadratics	12 min
A short proof	8 min

LEARNING OBJECTIVES

- Define increasing and decreasing on an interval.
- Read the intervals of monotonicity from a graph.
- Find them for a quadratic from its vertex.
- Prove a simple case from the definition.

TEXTBOOK

National	pp. 41–44
Cambridge	Core & Extended, pp. 198–215
Workbook	Exercise 10.1

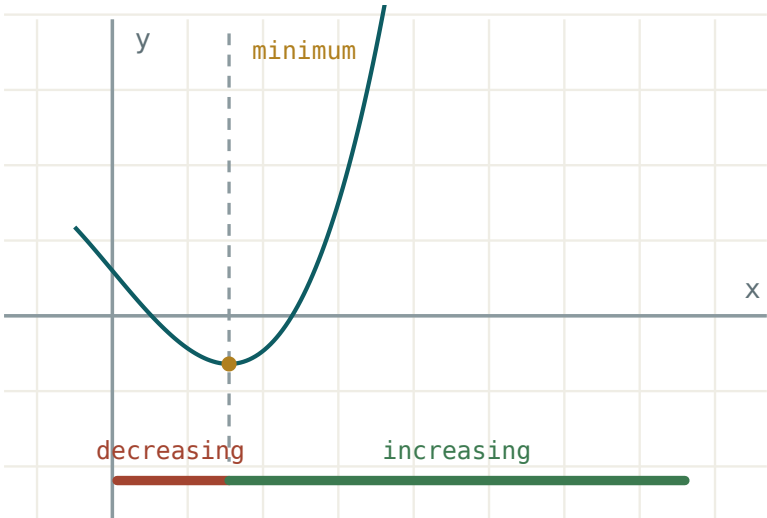
01

Explanation

The definition

f **increases** on I if $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$ for all $x_1, x_2 \in I$

f **decreases** on I if $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$



Uphill on one interval, downhill on another — read left to right, always.

ALWAYS “ON AN INTERVAL”, NEVER “AT A POINT”

A function is not increasing at $x = 2$; it increases *on* $[2, 5]$ or on $(-\infty, 1]$. Naming the interval is half the answer, and an answer without one earns no mark.

Reading a graph

THE CURVE	READING LEFT TO RIGHT	NAME
rises	y grows	increasing
falls	y shrinks	decreasing
flat	y unchanged	constant

The intervals change only at a **turning point** or where the function is undefined. So find those first, and the intervals are what lies between them.

Quadratics

A parabola has exactly one turning point — the vertex — so it has exactly two intervals of monotonicity.

A	LEFT OF THE VERTEX	RIGHT OF THE VERTEX
$a > 0$	decreasing	increasing
$a < 0$	increasing	decreasing

Example. $y = x^2 - 6x + 5$ has vertex at $x = 3$ and $a > 0$, so it decreases on $(-\infty, 3]$ and increases on $[3, +\infty)$.

THE VERTEX IS THE ONLY PLACE IT CAN TURN

Find $x = -\frac{b}{2a}$, look at the sign of a , and both intervals are written. No table of values is needed at all.

A short proof

Take $f(x) = 3x + 1$ and any $x_1 < x_2$:

$$f(x_2) - f(x_1) = 3(x_2 - x_1) > 0$$

Since $x_2 - x_1 > 0$, the difference is positive, so $f(x_2) > f(x_1)$: the function increases on the whole of $\mathbb{R}$.

THE PATTERN OF EVERY SUCH PROOF

Form $f(x_2) - f(x_1)$, factor out $(x_2 - x_1)$, and decide the sign of what is left. For $y = kx + b$ the answer is just the sign of k — which is why a line is increasing exactly when its gradient is positive.

02 Worked examples

EXAMPLE 1 Find the intervals of monotonicity of $y = x^2 - 6x + 5$.

① $x = \frac{6}{2} = 3$

The vertex.

② $a = 1 > 0$

③ Decreasing on $(-\infty, 3]$.

④ Increasing on $[3, +\infty)$.

ANSWER

Decreasing on $(-\infty, 3]$, increasing on $[3, +\infty)$

EXAMPLE 2 Find them for $y = -2x^2 + 8x - 1$.

① $x = -\frac{8}{-4} = 2$

② $a = -2 < 0$

③ Increasing on $(-\infty, 2]$, decreasing on $[2, +\infty)$.

ANSWER

Increasing on $(-\infty, 2]$, decreasing on $[2, +\infty)$

EXAMPLE 3 Prove that $f(x) = 3x + 1$ increases on $\mathbb{R}$.

① Take any $x_1 < x_2$.

② $f(x_2) - f(x_1) = 3x_2 + 1 - 3x_1 - 1$

③ $= 3(x_2 - x_1) > 0$

④ So $f(x_2) > f(x_1)$ — increasing.

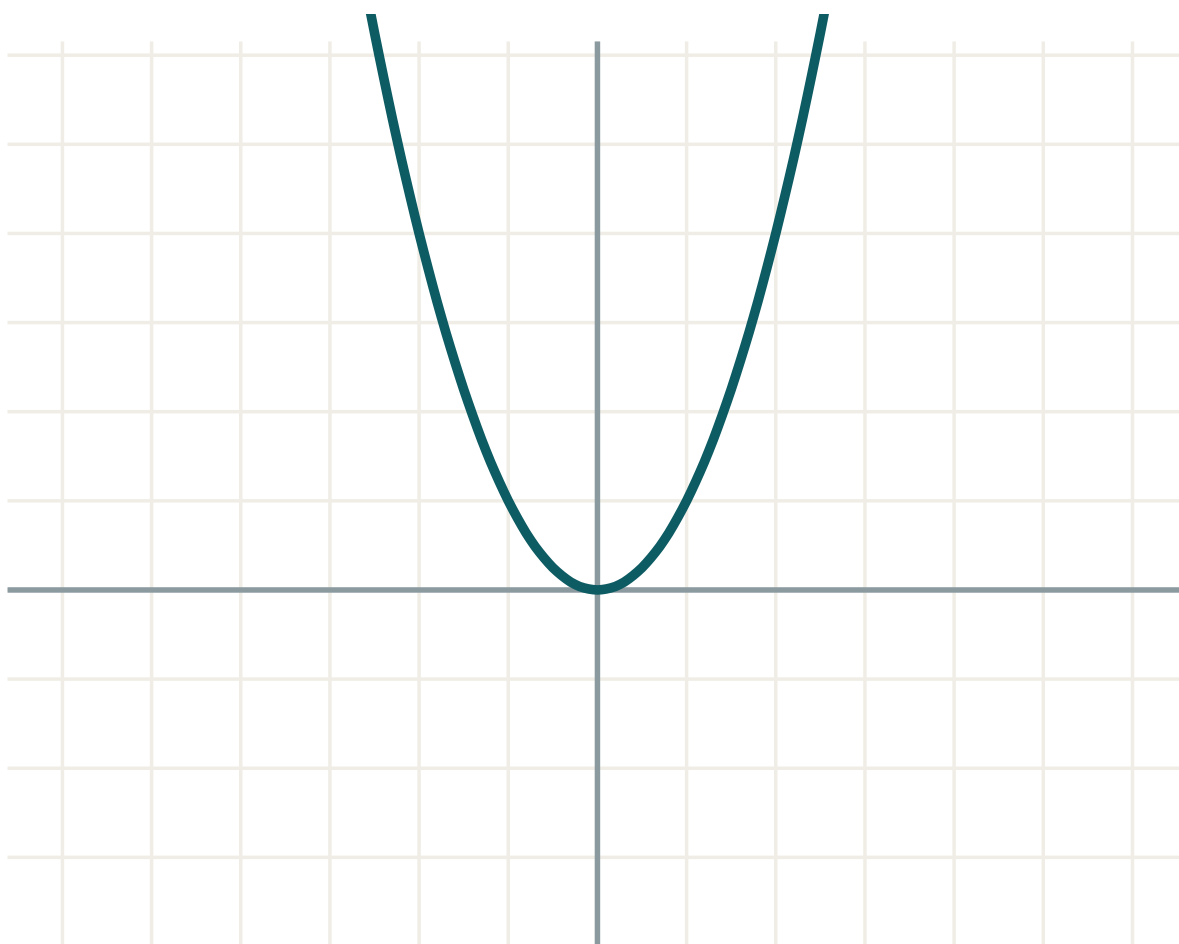
ANSWER

Increasing everywhere, since $k = 3 > 0$

03 Interactive model

Trace a finger along a drawn curve from left to right and let the class call out “up” or “down” — the definition follows the finger.

Where does it rise?



a 1

b 0

c 0

equation $y = f(x)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Increasing	O'suvchi	Возрастающая
Decreasing	Kamayuvchi	Убывающая
Monotonic	Monoton	Монотонная
Interval of increase	O'sish oralig'i	Промежуток возрастания
Turning point	Burilish nuqtasi	Точка поворота
Constant function	O'zgarmas funksiya	Постоянная функция
On an interval	Oraliqda	На промежутке
Definition	Ta'rif	Определение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Increasing means:

$$x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$$

$$f(x) > 0$$

the graph is above Ox

$$a > 0$$

Monotonicity is stated:

at a point

on an interval

for one value

never

 $y = x^2 - 6x + 5$ decreases on:

$$(-\infty, 3]$$

$$[3, +\infty)$$

all x

$$[0, 3]$$

For $a < 0$, right of the vertex the function:

increases

decreases

is constant

is undefined

$y = kx + b$ increases exactly when:

$b > 0$

$k > 0$

$k < 0$

always

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Is $y = 5x + 2$ increasing?

ANSWER Yes — $k = 5 > 0$

2. Is $y = -3x + 1$ increasing?

ANSWER No — decreasing

3. Vertex of $y = x^2 - 6x + 5$

ANSWER $x = 3$

4. Vertex of $y = -2x^2 + 8x - 1$

ANSWER $x = 2$

5. $y = x^2$ decreases on:

ANSWER $(-\infty, 0]$

6. $y = x^2$ increases on:

ANSWER $[0, +\infty)$

7. Is $y = 7$ increasing?

ANSWER No — constant

Medium · the standard question

7 PROBLEMS

1. Monotonicity of $y = x^2 - 6x + 5$

ANSWER Down on $(-\infty, 3]$, up on $[3, +\infty)$

2. Monotonicity of $y = -2x^2 + 8x - 1$

ANSWER Up on $(-\infty, 2]$, down on $[2, +\infty)$

3. Monotonicity of $y = (x + 4)^2 - 1$

ANSWER Down on $(-\infty, -4]$, up after

4. Monotonicity of $y = 3 - x^2$

ANSWER Up on $(-\infty, 0]$, down after

5. Prove $f(x) = 3x + 1$ increases

ANSWER $f(x_2) - f(x_1) = 3(x_2 - x_1) > 0$

6. Prove $f(x) = -x + 5$ decreases

ANSWER $f(x_2) - f(x_1) = -(x_2 - x_1) < 0$

7. How many turning points has a parabola?

ANSWER One

Hard · stretch and reason

7 PROBLEMS

1. Monotonicity of $y = \frac{1}{x}$ on $(0, +\infty)$

ANSWER Decreasing

2. Monotonicity of $y = \sqrt{x}$

ANSWER Increasing on $[0, +\infty)$

3. Monotonicity of $y = |x|$

ANSWER Down on $(-\infty, 0]$, up on $[0, +\infty)$

4. For which k is $y = (k - 2)x + 1$ decreasing?

ANSWER $k < 2$

5. Prove $f(x) = x^2$ increases on $[0, +\infty)$

ANSWER $x_2^2 - x_1^2 = (x_2 - x_1)(x_2 + x_1) > 0$

6. On which interval is $y = x^2 - 4x$ decreasing and negative?

ANSWER $(0, 2]$

7. Monotonicity of $y = -\frac{2}{x}$ on $(0, +\infty)$

ANSWER Increasing

07 Homework

Homework — 4 tasks

Every answer names an interval; a bare “increasing” earns nothing.

- Find the intervals of monotonicity of $y = x^2 - 10x + 3$.
- Find them for $y = -x^2 - 4x + 1$.
- Prove from the definition that $f(x) = 4x - 9$ increases on $\mathbb{R}$.

4. Sketch any function that increases on $(-\infty, -1]$, decreases on $[-1, 2]$ and increases on $[2, +\infty)$.

Even and odd functions

Two symmetries: one in the axis Oy, one about the origin — and most functions have neither.

LESSON 20–21

2 × 40 MIN

ALGEBRA 9, §11

EXTENSION BEYOND IGC

LESSON CLOCK

11'

18'

22'

20'

9'

The two definitions	11 min
The domain comes first	18 min
Testing	22 min
On the graph	20 min
Homework	9 min

LEARNING OBJECTIVES

- Test a function for evenness and oddness from $f(-x)$.
- Recognise the two symmetries on a graph.
- Check the domain for symmetry before testing anything else.
- Use parity to halve the work of sketching.

TEXTBOOK

National	pp. 45–50
Cambridge	Core & Extended, pp. 198–215
Workbook	Exercise 10.1

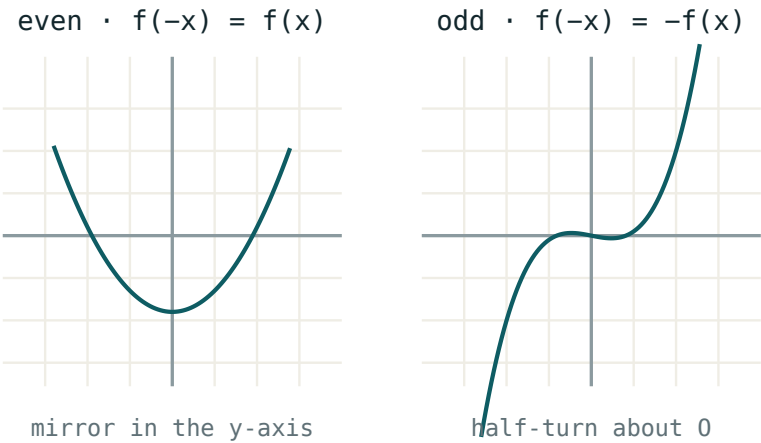
01

Explanation

The two definitions

f is **even** if $f(-x) = f(x)$ for every x of the domain

f is **odd** if $f(-x) = -f(x)$ for every x of the domain



Even — a mirror in Oy. Odd — a half-turn about the origin.

FUNCTION	$F(-X)$	VERDICT
x^2	x^2	even
x^3	$-x^3$	odd
$x^2 + 1$	$x^2 + 1$	even
$x + 1$	$-x + 1$	neither
$ x $	$ x $	even
$\frac{1}{x}$	$-\frac{1}{x}$	odd

WHERE THE NAMES COME FROM

x^n is even exactly when n is an even number, and odd exactly when n is odd. That is the whole origin of the two words — and a quick way to guess the answer before testing it.

The domain comes first

Both definitions say “for every x of the domain”, and they use $-x$. So $-x$ must itself be in the domain, or the question cannot even be asked.

a function can be even or odd only if its domain is **symmetric about 0**

FUNCTION	DOMAIN	SYMMETRIC?	VERDICT
x^2	all x	yes	test it
$\frac{1}{x}$	$x \neq 0$	yes	test it
$\sqrt{x}$	$[0, +\infty)$	no	neither, at once
$\sqrt{x-1}$	$[1, +\infty)$	no	neither, at once

CHECK THE DOMAIN BEFORE SUBSTITUTING

$\sqrt{x}$ is neither even nor odd — not because $\sqrt{-x}$ comes out wrong, but because $\sqrt{-x}$ does not exist. Saying so is the whole answer, and it takes one line.

Testing

THE THREE STEPS, EVERY TIME

1. Is the domain symmetric about 0 ? If not, stop: **neither**.
2. Compute $f(-x)$ and simplify.
3. Compare with $f(x)$ and with $-f(x)$.

Example. $f(x) = x^4 - 3x^2 + 2$. The domain is all of $\mathbb{R}$ ✓.

$$f(-x) = (-x)^4 - 3(-x)^2 + 2 = x^4 - 3x^2 + 2 = f(x)$$

So it is even. **Example.** $g(x) = x^3 - 4x$:

$$g(-x) = -x^3 + 4x = -(x^3 - 4x) = -g(x)$$

So it is odd. **Example.** $h(x) = x^2 + x$: $h(-x) = x^2 - x$, which is neither $h(x)$ nor $-h(x)$ — neither.

A POLYNOMIAL IS EASY TO READ OFF

All powers even (counting a constant as x^0) → even. All powers odd → odd. A mixture → neither. $x^4 - 3x^2 + 2$ is even by inspection; $x^3 - 4x$ is odd; $x^2 + x$ is neither.

On the graph

PARITY	SYMMETRY	WHAT IT SAVES YOU
even	mirror in Oy	plot $x \geq 0$, then reflect
odd	half-turn about O	plot $x \geq 0$, then rotate 180°
neither	none	nothing

An **odd** function whose domain contains 0 must have $f(0) = 0$: putting $x = 0$ in $f(-x) = -f(x)$ gives $f(0) = -f(0)$, so $2f(0) = 0$. That single check rules out many candidates instantly.

ONLY ONE FUNCTION IS BOTH

$f(x) = 0$ satisfies both definitions at once. Every other function is even, odd, or neither — never two of them.

02 Worked examples

EXAMPLE 1 Test $f(x) = x^4 - 3x^2 + 2$ for parity.

1 Domain $\mathbb{R}$ — symmetric ✓

2 $f(-x) = x^4 - 3x^2 + 2$
Even powers only.

3 $= f(x)$

4 Even.

ANSWER

Even

EXAMPLE 2 Test $g(x) = x^3 - 4x$.

1 Domain $\mathbb{R}$ ✓

2 $g(-x) = -x^3 + 4x$

3 $= -(x^3 - 4x) = -g(x)$

4 Odd.
Check: $g(0) = 0$ ✓

ANSWER

Odd

EXAMPLE 3 Test $h(x) = \sqrt{x} + 1$.

- ① Domain $[0, +\infty)$.
- ② -1 is not in the domain although 1 is.
Not symmetric.
- ③ Neither, and no substitution is needed.

ANSWER

Neither — the domain is not symmetric

EXAMPLE 4 Test $p(x) = \frac{x}{x^2 - 1}$.

- ① Domain $x \neq \pm 1$ — symmetric ✓
- ② $p(-x) = \frac{-x}{x^2 - 1}$
- ③ $= -p(x)$
- ④ Odd.

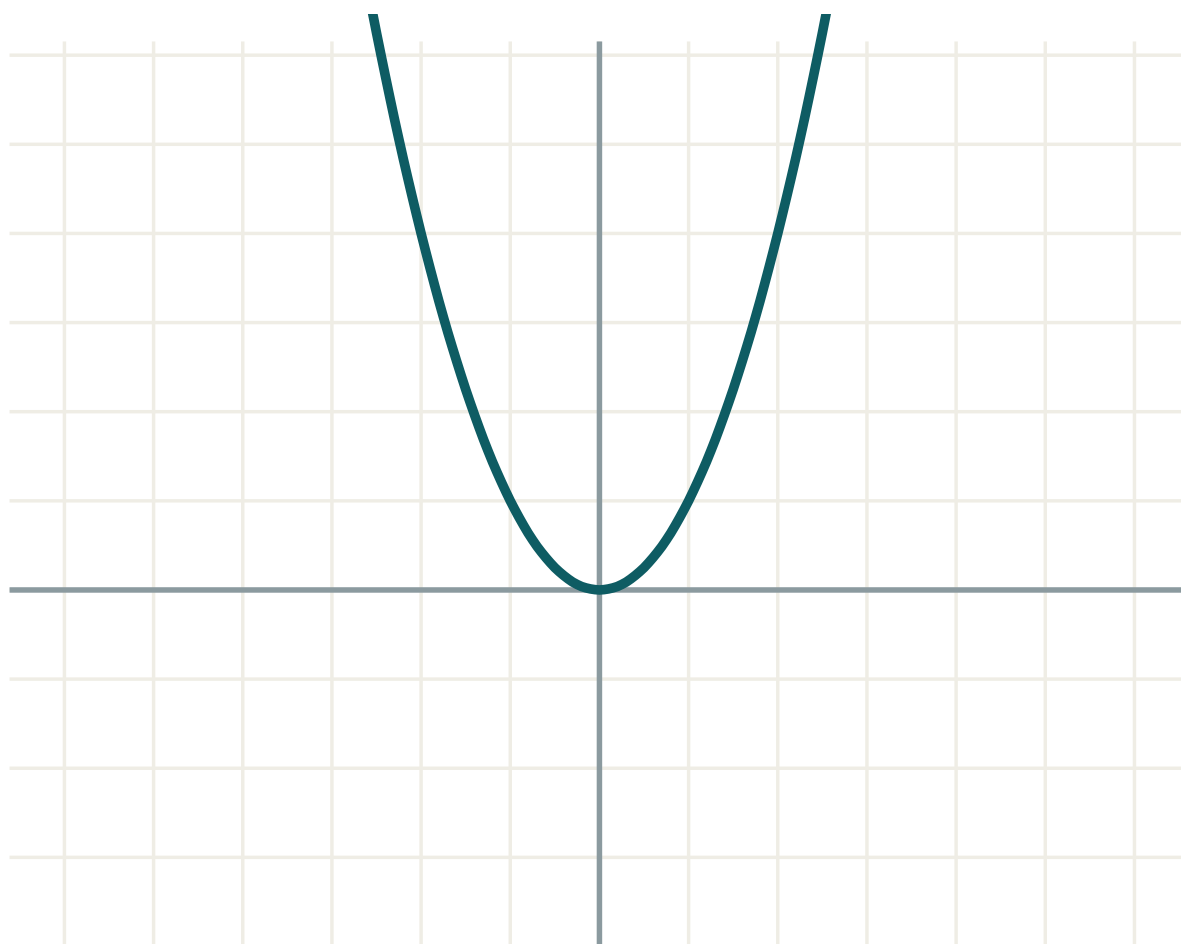
ANSWER

Odd

03 Interactive model

Fold a printed graph along Oy for even, and turn it upside down for odd — the class sees the two symmetries physically.

Two kinds of symmetry



a 1

b 0

c 0

equation $y = f(x)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Even function	Juft funksiya	Чётная функция
Odd function	Toq funksiya	Нечётная функция
Parity	Juft-toqlik	Чётность
Symmetric domain	Simmetrik soha	Симметричная область
Symmetry in Oy	Oy ga nisbatan simmetriya	Симметрия относительно Oy
Symmetry about O	O ga nisbatan simmetriya	Симметрия относительно начала
Neither	Na juft, na toq	Ни чётная, ни нечётная
Substitution	O'rniga qo'yish	Подстановка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Even means:

$$f(-x) = f(x)$$

$$f(-x) = -f(x)$$

$$f(x) > 0$$

$$f(0) = 0$$

Odd means:

$$f(-x) = f(x)$$

$$f(-x) = -f(x)$$

$$f(x) < 0$$

the powers are odd

Before testing you must check:

the range

that the domain is symmetric about 0

the roots

the vertex

$\sqrt{x}$ is:

even

odd

neither

both

$x^3 - 4x$ is:

even

odd

neither

both

An odd function with 0 in its domain has:

$f(0) = 1$

$f(0) = 0$

$f(0) > 0$

no value at 0

Which function is both even and odd?

x

x^2

$f(x) = 0$

none

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Parity of $y = x^2$

ANSWER Even

2. Parity of $y = x^3$

ANSWER Odd

3. Parity of $y = x + 1$

ANSWER Neither

4. Parity of $y = |x|$

ANSWER Even

5. Parity of $y = \frac{1}{x}$

ANSWER Odd

6. Parity of $y = 5$

ANSWER Even

7. Is the domain of $\sqrt{x}$ symmetric?

ANSWER No

Medium · the standard question

7 PROBLEMS

1. Parity of $y = x^4 - 3x^2 + 2$

ANSWER Even

2. Parity of $y = x^3 - 4x$

ANSWER Odd

3. Parity of $y = x^2 + x$

ANSWER Neither

4. Parity of $y = \frac{x}{x^2 - 1}$

ANSWER Odd

5. Parity of $y = \sqrt{x} + 1$

ANSWER Neither — domain

6. Parity of $y = x|x|$

ANSWER Odd

7. Parity of $y = x^5 + x^3 + x$

ANSWER Odd

Hard · stretch and reason

7 PROBLEMS

1. Parity of $y = \frac{x^2 + 1}{x^2 - 4}$

ANSWER Even

2. Parity of $y = \sqrt{4 - x^2}$

ANSWER Even

3. Parity of $y = x^3 + x^2$

ANSWER Neither

4. For which a is $y = x^3 + ax^2$ odd?

ANSWER $a = 0$

5. Show that the product of two odd functions is even

ANSWER $(-f)(-g) = fg$

6. Show that the sum of an even and an odd function is usually neither

ANSWER $f(-x) + g(-x) = f(x) - g(x)$

7. Parity of $y = \frac{1}{x^3} - x$

ANSWER Odd

07 Homework

Homework — 5 tasks

Begin every answer with the domain; parity questions are lost there more often than anywhere else.

1. Test $y = x^6 - x^2$ for parity.

2. Test $y = x^5 - 2x$ for parity.

3. Test $y = \sqrt{x + 2}$ for parity, giving the reason in one sentence.

4. Test $y = \frac{x^3}{x^2 + 1}$ for parity.

5. Explain in two sentences why an odd function defined at 0 must pass through the origin.

Equations and inequalities containing a power

Even powers lose the sign and need $\pm$; odd powers keep it and need nothing.

LESSON 22–23

2 × 40 MIN

ALGEBRA 9, §12

IGX 2.5

LESSON CLOCK

13'

20'

23'

20'

4'

Even against odd	13 min
Equations	20 min
Inequalities	23 min
Biquadratics	20 min
Homework	4 min

LEARNING OBJECTIVES

- Solve $x^n = a$ for even and for odd n .
- Solve inequalities of the form $x^n > a$ and $x^n < a$.
- Use substitution on a biquadratic equation.
- Reject the roots that the substitution forbids.

TEXTBOOK

National	pp. 51–55
Cambridge	Core & Extended, pp. 35–43
Workbook	Exercise 2.5

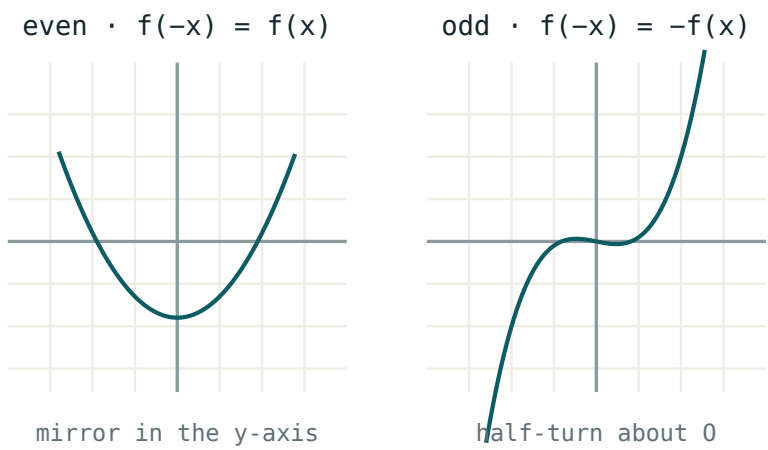
01

Explanation

Even against odd

The whole lesson turns on one fact: an even power destroys the sign, an odd power keeps it.

	EVEN N	ODD N
$(-2)^n$	positive	negative
range of x^n	$[0, +\infty)$	$\mathbb{R}$
graph	parabola-like, symmetric in Oy	rising, symmetric about O
$x^n = a$	two roots, one, or none	always exactly one



The two shapes — even powers fold, odd powers pass through.

THE CONSEQUENCE FOR SOLVING

$x^4 = 16$ has two solutions ± 2 ; $x^3 = 8$ has exactly one, 2 ; $x^4 = -16$ has none, and $x^3 = -8$ has one, -2 .
Read the parity of the exponent before anything else.

Equations

$x^{\{2k\}} = a: x = \pm \sqrt[2k]{a}$ if $a > 0$; $x = 0$ if $a = 0$; none if $a < 0$

$x^{\{2k+1\}} = a: x = \sqrt[2k+1]{a}$, always exactly one

EQUATION	SOLUTIONS
$x^2 = 49$	± 7
$x^4 = 81$	± 3
$x^4 = -81$	none
$x^3 = 27$	3
$x^5 = -32$	-2
$(x - 1)^4 = 16$	$x = 3, -1$

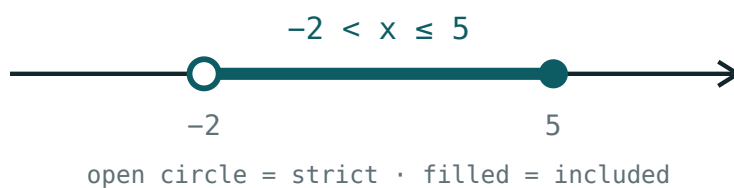
DO NOT LOSE THE NEGATIVE ROOT

$x^4 = 81$ gives $x = \pm 3$, not $x = 3$. And $(x - 1)^4 = 16$ gives $x - 1 = \pm 2$, so **both** $x = 3$ and $x = -1$.

Inequalities

For an even power the solution set is symmetric; for an odd power it is a single ray.

INEQUALITY	SOLUTION
$x^2 < 9$	$(-3, 3)$
$x^2 > 9$	$(-\infty, -3) \cup (3, +\infty)$
$x^4 \leq 16$	$[-2, 2]$
$x^2 < -4$	$\emptyset$
$x^2 > -4$	all x
$x^3 < 27$	$(-\infty, 3)$
$x^3 > -8$	$(-2, +\infty)$



Even powers give an interval about the origin or its complement; odd powers give one ray.

$x^2 < A$ IS A MODULUS STATEMENT

$x^2 < 9$ says $|x| < 3$, and $x^2 > 9$ says $|x| > 3$. Reading it that way makes the symmetry obvious and the brackets automatic.

Biquadratics

An equation $ax^4 + bx^2 + c = 0$ contains only even powers, so $t = x^2$ turns it into a quadratic.

$t = x^2 \geq 0$ — every negative root of the quadratic is rejected

Example. $x^4 - 5x^2 + 4 = 0$. With $t = x^2$: $t^2 - 5t + 4 = 0$, so $t = 1$ or $t = 4$. Both are non-negative, so $x^2 = 1$ gives $x = \pm 1$ and $x^2 = 4$ gives $x = \pm 2$ — four roots.

EQUATION	IN t	KEPT	ROOTS
$x^4 - 5x^2 + 4 = 0$	$t = 1, 4$	both	$\pm 1, \pm 2$
$x^4 - 3x^2 - 4 = 0$	$t = 4, -1$	$t = 4$	± 2
$x^4 + 5x^2 + 4 = 0$	$t = -1, -4$	neither	none

UNDO THE SUBSTITUTION, AND REJECT BEFORE YOU DO

Stopping at $t = 4$ answers a different question. And carrying $t = -1$ forward gives $x^2 = -1$, which has no real solution — say so rather than writing an imaginary answer in Grade 9.

02 Worked examples

EXAMPLE 1 Solve $x^4 = 81$ and $x^3 = -64$.

- 1

$x^4 = 81$: even power, $81 > 0$.
- 2

$x = \pm 3$
- 3

$x^3 = -64$: odd power.
- 4

$x = -4$
Exactly one root.

ANSWER

$x = \pm 3$ and $x = -4$

EXAMPLE 2 Solve $x^4 - 5x^2 + 4 = 0$.

- ① Let $t = x^2 \geq 0$.
- ② $t^2 - 5t + 4 = 0 \Rightarrow t = 1, 4$
- ③ Both are non-negative — keep both.
- ④ $x = \pm 1$ and $x = \pm 2$.

ANSWER

$$x = \pm 1, \pm 2$$

EXAMPLE 3 Solve $x^4 - 3x^2 - 4 = 0$.

- ① $t^2 - 3t - 4 = 0 \Rightarrow t = 4$ or $t = -1$
- ② $t = -1$ is rejected: $x^2 \geq 0$.
- ③ $x^2 = 4$
- ④ $x = \pm 2$

ANSWER

$$x = \pm 2$$

EXAMPLE 4 Solve $(x - 1)^4 \leq 16$.

$$\textcircled{1} \quad |x - 1| \leq 2$$

Even power, so a modulus.

$$\textcircled{2} \quad -2 \leq x - 1 \leq 2$$

$$\textcircled{3} \quad -1 \leq x \leq 3$$

ANSWER

$$[-1, 3]$$

03 **Interactive model**

Put $y = x^2$ and $y = x^3$ on the same axes and let the class say, for each horizontal line, how many times it cuts each curve.

Powers and their roots

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64**index n **3**power m **1**check: value³ **64**a¹ **64**

Raising the answer to the power 3 gives back a^1 — that is exactly what “root of degree 3” means.

04 **Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Power	Daraja	Степень
Even exponent	Juft ko'rsatkich	Чётный показатель
Odd exponent	Toq ko'rsatkich	Нечётный показатель
Biquadratic equation	Bikvadrat tenglama	Биквадратное уравнение
Substitution	O'rniga qo'yish	Подстановка
Root of degree n	n-darajali ildiz	Корень n-й степени
Extraneous root	Chet ildiz	Посторонний корень
Modulus	Modul	Модуль

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself** $x^4 = 16$ has: $x^3 = 8$ has: $x^4 = -16$ has:

$x^2 < 9$ means:

In $x^4 + bx^2 + c = 0$ the substitution needs:

$t = -1$ in a biquadratic gives:

$x^3 < 27$ gives:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up**7 PROBLEMS**

1. Solve $x^2 = 49$

ANSWER ± 7

2. Solve $x^3 = 27$

ANSWER 3

3. Solve $x^4 = 81$

ANSWER ± 3

4. Solve $x^5 = -32$

ANSWER -2

5. Solve $x^4 = -81$

ANSWER No real root

6. Solve $x^2 < 9$

ANSWER $(-3, 3)$

7. Solve $x^3 < 27$

ANSWER $(-\infty, 3)$ **Medium · the standard question****7 PROBLEMS**

1. Solve $x^4 - 5x^2 + 4 = 0$

ANSWER $\pm 1, \pm 2$

2. Solve $x^4 - 3x^2 - 4 = 0$

ANSWER ± 2

3. Solve $x^4 + 5x^2 + 4 = 0$

ANSWER No real root

4. Solve $(x - 1)^4 = 16$

ANSWER $x = 3, -1$

5. Solve $x^4 \leq 16$

ANSWER $[-2, 2]$

6. Solve $x^2 > -4$

ANSWER All x

7. Solve $x^3 > -8$

ANSWER $(-2, +\infty)$

Hard · stretch and reason

7 PROBLEMS

1. Solve $(x - 1)^4 \leq 16$

ANSWER $[-1, 3]$

2. Solve $x^6 - 9x^3 + 8 = 0$

ANSWER $x = 1, 2$

3. Solve $x^4 - 10x^2 + 9 < 0$

ANSWER $(-3, -1) \cup (1, 3)$

4. Solve $(2x + 1)^3 = -27$

ANSWER $x = -2$

5. For which a has $x^4 + a = 0$ a real root?

ANSWER $a \leq 0$

6. Solve $x^4 - 2x^2 - 8 \geq 0$

ANSWER $(-\infty, -2] \cup [2, +\infty)$

7. How many real roots has $x^4 - 4x^2 + 4 = 0$?

ANSWER Two — $x = \pm\sqrt{2}$

07 Homework

Homework — 5 tasks

For every biquadratic, write the condition $t \geq 0$ before solving.

1. Solve $x^4 = 625$ and $x^3 = -125$.

2. Solve $x^4 - 13x^2 + 36 = 0$.

3. Solve $x^4 + 3x^2 - 4 = 0$, saying which root of the quadratic you rejected and why.

4. Solve $(x + 2)^4 < 81$.

5. Explain in two sentences why $x^5 = a$ always has exactly one real solution.

Chapter exercises

— rational and irrational numbers

[Cambridge insert]

Which numbers can be written as a fraction, and which provably cannot.

LESSON 24–25

2 × 40 MIN

ALGEBRA 9, УПРАЖНЕНИЯ К ГЛАВЕ I

IGX 9.2

LESSON CLOCK

13'

20'

23'

20'

4'

The number sets	13 min
Recurring decimals	20 min
Surds	23 min
Why $\sqrt{2}$ is irrational	20 min
Homework	4 min

LEARNING OBJECTIVES

- Classify a number as natural, integer, rational or irrational.
- Convert a recurring decimal to a fraction.
- Simplify surds and rationalise a denominator.
- Follow the proof that $\sqrt{2}$ is irrational.

TEXTBOOK

National	pp. 56–62
Cambridge	Core & Extended, pp. 176–182
Workbook	Exercise 9.2

01

Explanation

The number sets

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$$

SET	SYMBOL	CONTAINS	EXAMPLE
natural	$\mathbb{N}$	counting numbers	1, 2, 3, ...
integers	$\mathbb{Z}$	with zero and negatives	..., -1, 0, 1, ...
rational	$\mathbb{Q}$	$\frac{p}{q}$ with $q \neq 0$	$\frac{3}{4}$, -2, 0.7, 0.333...
irrational	—	everything else real	$\sqrt{2}$, π , $\sqrt{3}$

THE DECIMAL TEST

A number is **rational** exactly when its decimal expansion terminates or recurs. 0.25 terminates; $0.333\dots$ recurs; $\sqrt{2} = 1.41421356\dots$ does neither, and never will.

Recurring decimals

Every recurring decimal is a fraction, and the conversion is three lines.

STEP	$0.272727\dots$
let	$x = 0.272727\dots$
multiply by 100	$100x = 27.272727\dots$
subtract	$99x = 27$
solve	$x = \frac{27}{99} = \frac{3}{11}$

The multiplier is 10 to the power of the **length of the repeating block**: 10 for one digit, 100 for two, 1000 for three.

$0.999\dots = 1$

Let $x = 0.999\dots$; then $10x = 9.999\dots$ and $9x = 9$, so $x = 1$. It is not an approximation — the two decimals are two names for the same number.

Surds

$$\sqrt{ab} = \sqrt{a} \cdot \sqrt{b} \quad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}} \quad (\sqrt{a})^2 = a \text{ for } a \geq 0$$

EXPRESSION	SIMPLIFIED
$\sqrt{50}$	$5\sqrt{2}$
$\sqrt{12} + \sqrt{27}$	$5\sqrt{3}$
$\sqrt{8} \times \sqrt{2}$	4
$\frac{1}{\sqrt{3}}$	$\frac{\sqrt{3}}{3}$
$\frac{2}{1 + \sqrt{3}}$	$\sqrt{3} - 1$

RATIONALISING USES THE CONJUGATE

Multiply top and bottom by $1 - \sqrt{3}$: the bottom becomes $(1 + \sqrt{3})(1 - \sqrt{3}) = 1 - 3 = -2$, a whole number. It is the same difference-of-squares trick as everywhere else.

$\sqrt{A} + \sqrt{B}$ IS NOT $\sqrt{A+B}$

$\sqrt{9} + \sqrt{16} = 3 + 4 = 7$, but $\sqrt{25} = 5$. The root of a sum is never the sum of the roots.

Why $\sqrt{2}$ is irrational

Suppose it were rational. Then $\sqrt{2} = \frac{p}{q}$ for whole numbers with no common factor.

STEP	REASON
$p^2 = 2q^2$	square both sides
p^2 is even	it equals $2q^2$
p is even	an odd square is odd
$p = 2k$, so $4k^2 = 2q^2$	substitute
$q^2 = 2k^2$, so q is even	the same argument
both even — contradiction	they had no common factor

WHAT THE PROOF REALLY SHOWS

Not that $\sqrt{2}$ is hard to compute, but that **no** fraction can equal it — however many decimal places you take. The same argument works for $\sqrt{3}$, $\sqrt{5}$ and $\sqrt{p}$ for any prime, and it is the oldest proof of its kind in mathematics.

02 Worked examples

EXAMPLE 1 Write $0.272727\dots$ as a fraction.

$$(1) \quad x = 0.272727\dots$$

$$(2) \quad 100x = 27.272727\dots$$

Two repeating digits.

$$(3) \quad 99x = 27$$

$$(4) \quad x = \frac{27}{99} = \frac{3}{11}$$

ANSWER

$$\frac{3}{11}$$

EXAMPLE 2 Simplify $\sqrt{12} + \sqrt{27}$.

$$\textcircled{1} \quad \sqrt{12} = 2\sqrt{3}$$

$$\textcircled{2} \quad \sqrt{27} = 3\sqrt{3}$$

$$\textcircled{3} \quad 2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$$

ANSWER

$$5\sqrt{3}$$

EXAMPLE 3 Rationalise $\frac{2}{1 + \sqrt{3}}$.

$$\textcircled{1} \quad \text{Multiply by } \frac{1 - \sqrt{3}}{1 - \sqrt{3}}.$$

$$\textcircled{2} \quad \text{bottom} = 1 - 3 = -2$$

$$\textcircled{3} \quad \frac{2(1 - \sqrt{3})}{-2}$$

$$\textcircled{4} \quad = \sqrt{3} - 1$$

ANSWER

$$\sqrt{3} - 1$$

EXAMPLE 4 Is $\sqrt{2} + \sqrt{8}$ rational or irrational?

① $\sqrt{8} = 2\sqrt{2}$

② $\sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$

③ $3\sqrt{2}$ is a rational multiple of an irrational number.

④ Irrational.

ANSWER

Irrational — it equals $3\sqrt{2}$

03 Interactive model

Ask the class to find a fraction equal to $\sqrt{2}$ on a calculator; every attempt fails, and the proof explains why it must.

Roots, exact and approximate

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64**

index n **3**

power m **1**

check: value³ **64**

a^1 **64**

Raising the answer to the power 3 gives back a^1 — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Natural number	Natural son	Натуральное число
Integer	Butun son	Целое число
Rational number	Ratsional son	Рациональное число
Irrational number	Irratsional son	Иррациональное число
Real number	Haqiqiy son	Действительное число
Recurring decimal	Davriy o'nli kasr	Периодическая дробь
Surd	Irratsional ildiz	Радикал
Rationalising	Maxrajni ratsionallashtirish	Освобождение от иррациональности
Proof by contradiction	Teskarisini faraz qilish	Доказательство от противного
Set notation	To'plam belgisi	Обозначение множеств

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A rational number is one that:

has a decimal

can be written as $\frac{p}{q}$

is positive

is whole

A number is rational exactly when its decimal:

is short

terminates or recurs

has no zeros

is infinite

$0.999\dots$ equals:

0.99

just under 1

1

$\frac{9}{10}$

$\sqrt{50}$ simplifies to:

$\sqrt{9} + \sqrt{16}$ equals:

To rationalise $\frac{2}{1 + \sqrt{3}}$ multiply by:

The proof that $\sqrt{2}$ is irrational works by:

computing decimals

contradiction

induction

a graph

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Is -3 an integer?

ANSWER Yes

2. Is 0.25 rational?

ANSWER Yes

3. Is $\sqrt{2}$ rational?

ANSWER No

4. Is $\sqrt{9}$ rational?

ANSWER Yes — it is 3

5. $0.333\dots$ as a fraction

ANSWER $\frac{1}{3}$

6. $\sqrt{50}$ simplified

ANSWER $5\sqrt{2}$

7. $\sqrt{8} \times \sqrt{2}$

ANSWER 4

Medium · the standard question

7 PROBLEMS

1. $0.272727\dots$ as a fraction

ANSWER $\frac{3}{11}$

2. $0.4545\dots$ as a fraction

ANSWER $\frac{5}{11}$

3. $\sqrt{12} + \sqrt{27}$

ANSWER $5\sqrt{3}$

4. $\sqrt{18} - \sqrt{8}$

ANSWER $\sqrt{2}$

5. Rationalise $\frac{1}{\sqrt{3}}$

ANSWER $\frac{\sqrt{3}}{3}$

6. Rationalise $\frac{2}{1 + \sqrt{3}}$

ANSWER $\sqrt{3} - 1$

7. Is $\sqrt{2} + \sqrt{8}$ rational?

ANSWER No — $3\sqrt{2}$

Hard · stretch and reason

7 PROBLEMS

1. $0.1666\dots$ as a fraction

ANSWER $\frac{1}{6}$

2. Rationalise $\frac{\sqrt{5}}{\sqrt{5}-1}$

ANSWER $\frac{5+\sqrt{5}}{4}$

3. Is $(\sqrt{2}+1)(\sqrt{2}-1)$ rational?

ANSWER Yes — it is 1

4. Simplify $(\sqrt{3}+\sqrt{2})^2$

ANSWER $5+2\sqrt{6}$

5. Prove that $\sqrt{3}$ is irrational

ANSWER The same contradiction, with 3

6. Give an irrational number between 2 and 3

ANSWER $\sqrt{5} \approx 2.236$

7. Is the sum of two irrationals always irrational?

ANSWER No — $\sqrt{2} + (-\sqrt{2}) = 0$

07 Homework

Homework — 5 tasks

Give exact answers; a decimal is not a simplification of a surd.

1. Write $0.636363\dots$ and $0.8333\dots$ as fractions.

2. Simplify $\sqrt{75} - \sqrt{12}$.

3. Rationalise $\frac{3}{2 - \sqrt{2}}$.
4. Decide, with a reason, whether $(\sqrt{5} + 2)(\sqrt{5} - 2)$ is rational.
5. Write out the proof that $\sqrt{2}$ is irrational in your own words.

Control work 2, and work on the mistakes

The properties of functions in one paper, and the whole of Chapter I as a single map.

LESSON 26–27

2 × 40 MIN

ALGEBRA 9, NAZORAT ISHI 2

IGX 9–10 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Find a domain, a monotonicity interval and a parity under time.
- Solve an equation and an inequality containing a power.
- Classify each lost mark and rewrite the solution in full.
- Draw Chapter I as one map before the next chapter begins.

TEXTBOOK

National	pp. 60–67
Cambridge	Core & Extended, pp. 176–228
Workbook	Control paper A2

01

Explanation

The paper — 30 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Find the domain of $y = \frac{\sqrt{x+2}}{x-3}$	5	L17–18
2	Find the domain of $y = \sqrt{x^2 - x - 6}$	5	L17–18
3	Give the intervals of monotonicity of $y = x^2 + 6x - 1$	4	L19
4	Test $y = x^4 - x^2$ and $y = x^3 + x$ for parity	6	L20–21
5	Solve $x^4 - 13x^2 + 36 = 0$	6	L22–23
6	Simplify $\sqrt{32} - \sqrt{8}$ and rationalise $\frac{1}{2 + \sqrt{3}}$	4	L24–25

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for combining **both** conditions; Q2 one for the square brackets; Q3 one for naming the intervals rather than saying “increasing”; Q4 one for checking the domain first; Q5 one for undoing the substitution. Five of the thirty marks are habits.

Naming the slip

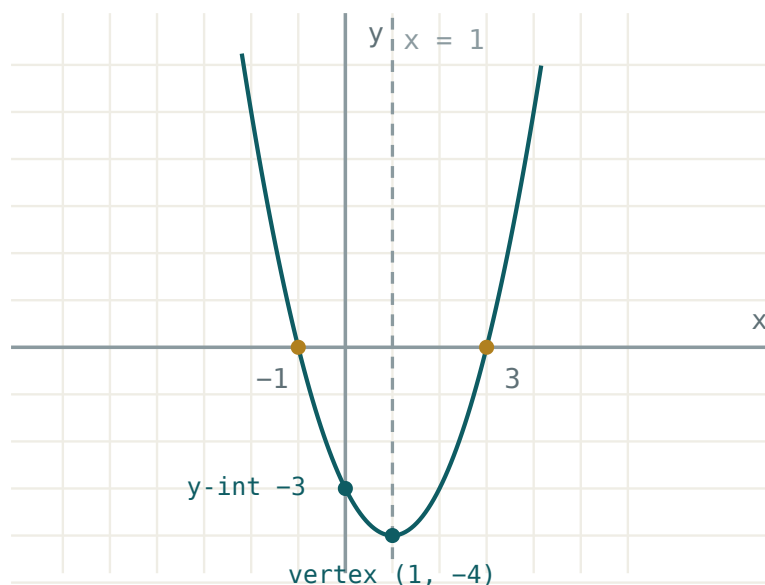
SLIP	WHAT IT LOOKS LIKE	THE FIX
one condition only	$[-2, +\infty)$	the denominator forbids 3 too
wrong side of the roots	$[-2, 3]$	outside for $\geq 0, a > 0$
no interval named	“increasing”	on $[-3, +\infty)$
parity without the domain	substituting into $\sqrt{x}$	check symmetry first
substitution not undone	$t = 4, 9$	$x = \pm 2, \pm 3$
negative t kept	$x^2 = -1$	reject it and say so
surd left unsimplified	$\sqrt{32}$	$4\sqrt{2}$

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

Chapter I as one map

Six boxes, and the links written as sentences:

- **a and D** → **the picture** — “direction, and how many roots”
- **completing the square** → **the vertex** — “and therefore the least or greatest value”
- **the roots** → **the sign** — “ a 's sign outside, the opposite between”
- **factors** → **the number line** — “mark, alternate, read”
- **denominators and roots** → **the domain** — “the only two things that forbid”
- **$f(-x)$** → **parity** — “but only if the domain is symmetric”



One picture, and five of the six boxes read off it.

LOOKING FORWARD

Chapter II takes the same quadratic and puts **two** of them together: systems of second-degree equations, systems of second-degree inequalities, and the proof of simple inequalities. Nothing new is needed — only what is on this map, used twice at once.

02 Worked examples

EXAMPLE 1 Model answer, Q1: the domain of $y = \frac{\sqrt{x+2}}{x-3}$.

① $x + 2 \geq 0 \Rightarrow x \geq -2$
The root.

② $x - 3 \neq 0 \Rightarrow x \neq 3$
The denominator.

③ Both must hold.

④ $[-2, 3) \cup (3, +\infty)$

ANSWER

$[-2, 3) \cup (3, +\infty)$

EXAMPLE 2 Model answer, Q4: test $y = x^4 - x^2$ and $y = x^3 + x$.

① Both domains are $\mathbb{R}$ — symmetric ✓

② $(-x)^4 - (-x)^2 = x^4 - x^2$ — even.
All powers even.

③ $(-x)^3 + (-x) = -x^3 - x = -(x^3 + x)$ — odd.
All powers odd.

④ Even, and odd, respectively.

ANSWER

The first is even, the second odd

EXAMPLE 3 Model answer, Q5: solve $x^4 - 13x^2 + 36 = 0$.

① Let $t = x^2 \geq 0$.

② $t^2 - 13t + 36 = 0 \Rightarrow t = 4, 9$

③ Both non-negative — keep both.

④ $x = \pm 2$ and $x = \pm 3$.

Four roots.

ANSWER

$x = \pm 2, \pm 3$

03 Interactive model

Work Q1 twice on the board — once with only the root condition — and let the class find the value that was wrongly admitted.

Chapter I in twelve questions

$y = ax^2 + bx + c$ is quadratic when:

$a \neq 0$

$b \neq 0$

$c \neq 0$

always

The vertex of $y = (x - 4)^2 - 1$:

$(4, -1)$

$(-4, -1)$

$(4, 1)$

$(1, 4)$

$D < 0$ and $a > 0$ mean the trinomial is:

always negative

always positive

sometimes zero

undefined

For $a > 0$, negative values occur:

outside the roots

between them

nowhere

everywhere

A double zero makes the sign:

flip

stay

vanish

invert

A denominator zero is drawn:

filled

open

either

not at all

The domain of $\sqrt{x - 4}$:

$x > 4$

$x \geq 4$

$x \leq 4$

all x

The domain of $\sqrt{x-4}$:

$x \geq 4$

$x > 4$

$x \neq 4$

all x

Monotonicity is stated:

at a point

on an interval

for one value

never

$x^3 - 4x$ is:

even

odd

neither

both

$x^4 = 16$ has:

one root

two roots

four

none

$\sqrt{32}$ simplifies to:

$$2\sqrt{8}$$

$$4\sqrt{2}$$

$$16\sqrt{2}$$

$$8\sqrt{2}$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Domain	Aniqlanish sohasi	Область определения
Monotonicity	Monotonlik	Монотонность
Parity	Juft-toqlik	Чётность
Biquadratic	Bikvadrat	Биквадратное
Surd	Irratsional ildiz	Радикал
Concept map	Tushunchalar xaritasi	Карта понятий
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 needs how many conditions?

one

two

three

none

Q3's answer must include:

the vertex only

the named intervals

the roots

a sketch

Q4 begins by checking:

$f(0)$

that the domain is symmetric

the roots

the range

Chapter II puts together:

- one quadratic
- two at once — systems
- trigonometry
- progressions

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Domain of $y = \sqrt{x + 2}$

ANSWER $[-2, +\infty)$

2. Domain of $y = \frac{1}{x - 3}$

ANSWER $x \neq 3$

3. Vertex of $y = x^2 + 6x - 1$

ANSWER $(-3, -10)$

4. Parity of $y = x^4 - x^2$

ANSWER Even

5. Parity of $y = x^3 + x$

ANSWER Odd

6. $\sqrt{32}$ simplified

ANSWER $4\sqrt{2}$

7. $\sqrt{8}$ simplified

ANSWER $2\sqrt{2}$

Medium · the standard question

7 PROBLEMS

1. Domain of $y = \frac{\sqrt{x+2}}{x-3}$

ANSWER $[-2, 3) \cup (3, +\infty)$

2. Domain of $y = \sqrt{x^2 - x - 6}$

ANSWER $(-\infty, -2] \cup [3, +\infty)$

3. Monotonicity of $y = x^2 + 6x - 1$

ANSWER Down on $(-\infty, -3]$, up after

4. Solve $x^4 - 13x^2 + 36 = 0$

ANSWER $\pm 2, \pm 3$

5. $\sqrt{32} - \sqrt{8}$

ANSWER $2\sqrt{2}$

6. Rationalise $\frac{1}{2 + \sqrt{3}}$

ANSWER $2 - \sqrt{3}$

7. Solve $x^4 = 16$

ANSWER ± 2

Hard · stretch and reason

7 PROBLEMS

1. Domain of $y = \frac{\sqrt{x^2 - 4}}{x - 3}$

ANSWER $(-\infty, -2] \cup [2, 3) \cup (3, +\infty)$

2. Parity of $y = \frac{x^4 + 1}{x^2}$

ANSWER Even

3. Solve $x^4 - 5x^2 - 36 = 0$

ANSWER ± 3

4. Monotonicity of $y = -x^2 + 8x$

ANSWER Up on $(-\infty, 4]$, down after

5. Simplify $(2 + \sqrt{3})(2 - \sqrt{3})$

ANSWER 1

6. Domain of $y = \frac{1}{\sqrt{x^2 - x - 6}}$

ANSWER $(-\infty, -2) \cup (3, +\infty)$

7. For which k is $y = x^2 + kx + 1$ decreasing on $(-\infty, 2]$?

ANSWER $k = -4$

07 Homework

Homework — 4 tasks

Bring the six-box map to the first lesson of Chapter II.

1. Rewrite in full every control-work question that lost a mark, naming the slip in the margin.
2. Complete the six-box concept map with every link written as a sentence.
3. Find the domain of $y = \frac{\sqrt{5 - x}}{x + 1}$.

4. Write your target for Chapter II in one checkable sentence, and date it.